Journal of Imaging Informatics in Medicine
https://doi.org/10.1007/s10278-026-02249-8

A Clinically Grounded Review of Medical Image Classification: Quantitative Insights into CNNs, Vision Transformers, and Hybrid CNN-ViT Models

Hamza Hussaini[1] · **Shahana Bano**[1] · **Eyad Elyan**[1] · **Carlos Francisco Moreno-Garcia**[1]

Received: 5 June 2026 / Revised: 20 August 2026 / Accepted: 21 August 2026
© The Author(s) 2026

Abstract

Medical image classification has advanced substantially with convolutional neural networks (CNNs), Vision Transformers (ViTs), and hybrid CNN–ViT architectures, yet clinical translation remains limited by dataset dependency, inconsistent evaluation practices, and insufficient external validation. This review provides a clinically grounded comparative synthesis of these model families across diverse medical imaging modalities. A structured literature review was conducted across PubMed, IEEE Xplore, Scopus, and Web of Science for studies published between 2016 and 2025. Following predefined eligibility criteria, 81 studies were included in the qualitative review, of which 74 contributed to a dataset-aware descriptive quantitative aggregation. The quantitative synthesis was therefore restricted to descriptive aggregation; a formal meta-analysis was not performed because of substantial methodological heterogeneity and insufficient reporting of study-level variance information across the included studies. CNN-based models demonstrated the most consistent performance, achieving the highest weighted accuracy (0.934) and weighted recall (0.901). ViT-based models achieved competitive weighted accuracy (0.906) and recall (0.893), particularly for OCT and X-ray imaging, but appeared more sensitive to dataset scale and quality. Hybrid CNN–ViT models achieved a weighted accuracy of 0.814 and weighted recall of 0.698, with the greatest performance variability. Only 10 of the 81 reviewed studies (12.3%) reported independent external validation, while calibration and other clinically relevant evaluation measures were inconsistently reported. CNNs provide a robust baseline for medical image classification, whereas ViT- and hybrid-based architectures offer complementary strengths under appropriate data and training conditions. However, limited external validation and inconsistent reporting indicate that strong retrospective performance should not be interpreted as evidence of clinical readiness. Future research should prioritise externally validated, interpretable, and clinically deployable AI systems supported by standardised evaluation practices.

Keywords Medical image classification · Convolutional Neural Networks (CNNs) · Vision Transformers (ViTs) · Hybrid CNN–ViT models · Descriptive quantitative aggregation · Clinical translation

Introduction

Medical image classification has become an important component of computer-aided diagnosis by supporting automated interpretation of complex imaging data for clinical decision-making [1]. Recent advances in deep learning, particularly convolutional neural networks (CNNs), Vision Transformers (ViTs), and hybrid CNN–ViT architectures, have substantially improved representation learning across imaging modalities including computed tomography (CT), X-ray, optical coherence tomography (OCT), ultrasound, and microscopic imaging [2]. Despite these advances, important challenges remain regarding generalisation, dataset dependency, interpretability, external validation, and reliable clinical deployment.

CNN-based models remain widely adopted owing to their effectiveness in learning local image representations and their comparatively stable performance under heterogeneous data conditions [3]. In contrast, ViT-based models employ

✉ Hamza Hussaini
h.hussaini@rgu.ac.uk

Shahana Bano
s.bano@rgu.ac.uk

Eyad Elyan
e.elyan@rgu.ac.uk

Carlos Francisco Moreno-Garcia
c.moreno-garcia@rgu.ac.uk

[1] Computing, Robert Gordon University, Garthdee Road, Aberdeen AB10 7QB, UK

Published online: 03 September 2026

self-attention mechanisms to capture global contextual information [4], although their performance often depends on larger and well-curated datasets. Hybrid CNN–ViT architectures aim to combine local and global representation learning but may remain sensitive to dataset characteristics, architectural design, and optimisation strategy.

Although numerous review studies have examined individual model families, most remain primarily descriptive and architecture-focused. Comparatively limited attention has been given to descriptive quantitative aggregation, dataset-scale sensitivity, performance variability across imaging modalities, and the broader implications for clinical translation. In this review, the term *clinically grounded* refers to an evaluation that goes beyond predictive performance, including robustness, interpretability, external validation, deployment feasibility, workflow integration, and potential regulatory relevance.

To address these gaps, this review provides a structured synthesis of CNN-, ViT-, and hybrid CNN–ViT-based medical image classification studies published between 2016 and 2025. The review combines qualitative analysis with a dataset-aware descriptive aggregation framework incorporating modality-wise aggregation, weighted performance summarisation, and variability analysis under heterogeneous evaluation settings. In addition to comparing model families, the review examines recurring methodological limitations and challenges related to data efficiency, interpretability, external validation, and clinical robustness.

Accordingly, RQ1 addresses the identified need for systematic cross-architecture comparison across heterogeneous imaging modalities and clinical applications, whereas RQ2 addresses the methodological and translational factors underlying reported performance, generalisability, interpretability, and clinical translation.

The following research questions guide the review:

1. **RQ1:** How do CNN-, Vision Transformer (ViT)-, and hybrid CNN–ViT-based architectures compare in medical image classification across different imaging modalities and clinical applications?
2. **RQ2:** What methodological factors influence their reported performance, generalisability, interpretability, and clinical translation?

The main contributions of this work are summarised as follows:

1. **Comprehensive review:** A structured review of 81 studies on CNN-, ViT-, and hybrid CNN–ViT architectures spanning diverse medical imaging modalities and clinical applications.
2. **Dataset-aware descriptive aggregation:** A descriptive aggregation of performance trends across 74 studies reporting comparable evaluation metrics, using modality-wise aggregation, weighted performance summarisation, and variability analysis.
3. **Clinical and methodological analysis:** Identification of recurring methodological challenges that affect clinical translation, including dataset dependence, limited external validation, inconsistent evaluation practices, interpretability, and reporting quality.
4. **Future research priorities:** Identification of key directions for developing more robust, interpretable, externally validated, and clinically deployable medical image classification systems.

Figure 1 presents the conceptual framework underpinning this review, illustrating the relationships among CNNs, Vision Transformers, hybrid CNN–ViT architectures, their clinical application domains, and the associated methodological and translational challenges. Table 1 then compares the present review with recent survey articles, highlighting its distinctive scope, methodological framework, and contributions.

Search Strategy, Study Selection, and Quality Assessment

This review adopted a structured systematic-style methodology guided by the PRISMA 2020 reporting recommendations [9, 10], combined with a dataset-aware descriptive aggregation framework. Although the review was not designed as a formal meta-analysis, the literature identification, screening, eligibility assessment, and reporting processes followed established systematic review principles. The completed PRISMA 2020 checklist, developed in accordance with the PRISMA 2020 reporting guideline [9], is provided in the Supplementary Material (Table 16). The review methodology was designed to address the research questions outlined in "Introduction" section.

A literature search was conducted across PubMed, IEEE Xplore, Scopus, and Web of Science to identify studies published between 2016 and 2025. The year 2016 was selected as the starting point because it marks the beginning of widespread adoption of deep learning for medical image classification following the rapid clinical translation of modern convolutional neural network architectures. The search period also captures the subsequent emergence of Vision Transformers and hybrid CNN–ViT models. Representative search terms included "medical image classification", "deep learning", "CNN", "ViT", and "hybrid CNN–ViT", combined using Boolean operators. The complete database-specific search strategies, search strings, and applied filters are presented in Table 2. Manual reference screening and backward snowballing were additionally performed to improve literature coverage.

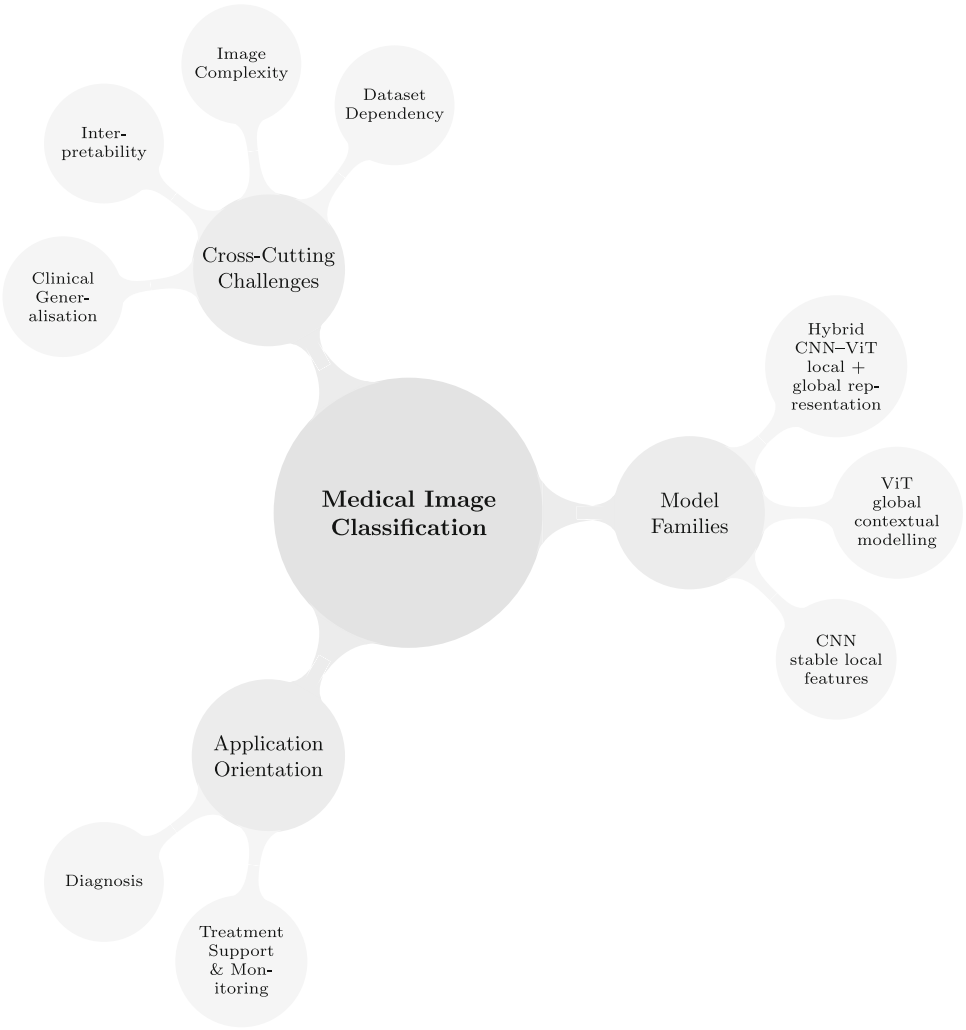


Fig. 1 Conceptual taxonomy adopted in this review, linking model families, application orientation, and cross-cutting challenges in medical image classification

Studies were included if they satisfied all of the following criteria:

1. investigated supervised medical image classification,
2. employed CNN-, ViT-, or hybrid CNN–ViT architectures,
3. reported at least one quantitative performance metric,
4. were published between 2016 and 2025 in Q1 journals or A/A*-ranked conferences, and
5. provided sufficient methodological information to support qualitative comparison.

Journal and conference rankings were verified using the SCImago Journal Rank (SJR)[1] and the CORE Conference Rankings,[2] respectively. Restricting inclusion to Q1 journals and A/A*-ranked conferences was intended to improve methodological consistency, reporting quality, and reproducibility of the comparative synthesis rather than to imply lower scientific quality of studies published in other venues.

Studies were excluded if they:

1. focused primarily on image segmentation, detection, registration, or image generation,
2. lacked experimental validation,
3. were review articles, editorials, or abstracts,
4. did not report quantitative evaluation results, or
5. fell outside the scope of medical image classification.

Following duplicate removal, titles and abstracts were screened against the eligibility criteria, followed by full-text assessment of potentially eligible studies. Screening was conducted by the first author and reviewed by the supervisory authors, with disagreements resolved through consensus. Formal inter-rater reliability statistics, such as Cohen's κ, were therefore not calculated, as independent parallel screening by two reviewers was not performed throughout the selection process; this is acknowledged as a methodological

[1] https://www.scimagojr.com/

[2] https://portal.core.edu.au/conf-ranks/

Table 1 Comparison of this review with recent survey studies on medical image classification

Study	Year	Model scope	Quantitative analysis	Generalisation analysis	Explainability	Clinical integration	Key focus/limitation
[5]	2023	ViT-focused	Descriptive	Limited	Discussed	Not addressed	Focused on transformer applications; limited CNN and hybrid comparison
[6]	2024	CNN vs ViT	Qualitative	Partial	Partial	Not addressed	Architectural comparison without quantitative synthesis
[7]	2025	CNN only	Descriptive	Limited	Covered	Limited	Focused on CNN methods without cross-model comparison
[8]	2025	Hybrid CNN–ViT	Benchmarking-based	Not explicit	Limited	Radiology-focused	Focused primarily on hybrid architectures and efficiency
This Review	–	**CNN + ViT + Hybrid**	**Dataset-aware descriptive aggregation**	**Explicit dataset dependency analysis**	**Context-aware attention perspective**	**Clinical workflow integration**	**Integrates descriptive aggregation, variability analysis, and clinical translation perspective**

Table 2 Database-specific representative search strategies and eligibility filters used for literature retrieval. Electronic database searches were complemented by manual reference screening and backward snowballing

Database	Representative search strategy	Period	Eligibility filters
PubMed	("medical image classification" OR "medical image analysis" OR "medical imaging") AND ("deep learning" OR "artificial intelligence" OR "machine learning") AND ("convolutional neural network" OR CNN OR "vision transformer" OR "vision transformers" OR ViT OR "hybrid CNN-ViT" OR "hybrid CNN-ViT")	2016–2025	English; peer-reviewed journals and conferences
IEEE Xplore	("medical image classification" OR "medical image analysis" OR "medical imaging") AND ("convolutional neural network" OR CNN OR "vision transformer" OR "vision transformers" OR ViT OR "hybrid CNN-ViT" OR "hybrid CNN-ViT")	2016–2025	English; peer-reviewed journals and conferences
Scopus	TITLE-ABS-KEY(("medical image classification" OR "medical image analysis" OR "medical imaging") AND ("convolutional neural network" OR CNN OR "vision transformer" OR "vision transformers" OR ViT OR "hybrid CNN-ViT" OR "hybrid CNN-ViT"))	2016–2025	English; peer-reviewed journals and conferences
Web of Science	TS=(("medical image classification" OR "medical image analysis" OR "medical imaging") AND ("convolutional neural network" OR CNN OR "vision transformer" OR "vision transformers" OR ViT OR "hybrid CNN-ViT" OR "hybrid CNN-ViT"))	2016–2025	English; SCI/SCIE-indexed journals and conferences

limitation. The complete study selection process is summarised in the PRISMA flow diagram (Fig. 2).

Following screening, 81 studies were included in the qualitative review. Of these, 74 studies reporting comparable quantitative performance metrics contributed to the weighted performance comparison and accuracy variability analyses, while 57 studies reporting recall contributed to the recall variability analysis.

A structured methodological quality assessment was undertaken to identify recurring strengths, limitations, and potential sources of bias across the included studies. The review considered methodological characteristics including dataset transparency, dataset scale, preprocessing description, train–validation–test separation, external validation, reporting completeness, explainability analysis, and reproducibility indicators, where reported. Particular attention was given to factors affecting clinical reliability and generalisability, including reliance on single datasets, limited external validation, and incomplete reporting of clinically relevant evaluation metrics. The assessment criteria are summarised in Appendix Table 15 and were applied descriptively rather than as a numerical quality score.

QUADAS-2 was developed for assessing the risk of bias and applicability of diagnostic accuracy studies, whereas PROBAST was developed for prediction model studies [11, 12]. Because the objective of the present review was methodological comparison of deep learning architectures rather than evaluation of diagnostic accuracy or prediction model performance, these tools were not considered appropriate. Consequently, a review-specific methodological quality assessment framework was adopted to evaluate reporting quality and potential sources of bias relevant to medical image classification studies.

A formal meta-analysis was not performed because most included studies did not report the study-level variance statistics (e.g., standard errors, confidence intervals, confusion matrices, or effect-size variance estimates) required for meta-analytic pooling. Furthermore, substantial heterogeneity in imaging modalities, datasets, evaluation protocols, and reporting standards limited the appropriateness of formal inferential synthesis. Instead, a structured dataset-aware descriptive aggregation strategy was adopted, as discussed in the "Discussion and Key Insights" section. Dataset-size aggregation was first performed within studies, followed by paper-level aggregation to enable comparative analysis while reducing the influence of exceptionally large datasets. Consequently, the findings should be interpreted as a structured descriptive aggregation rather than a formal statistical meta-analysis.

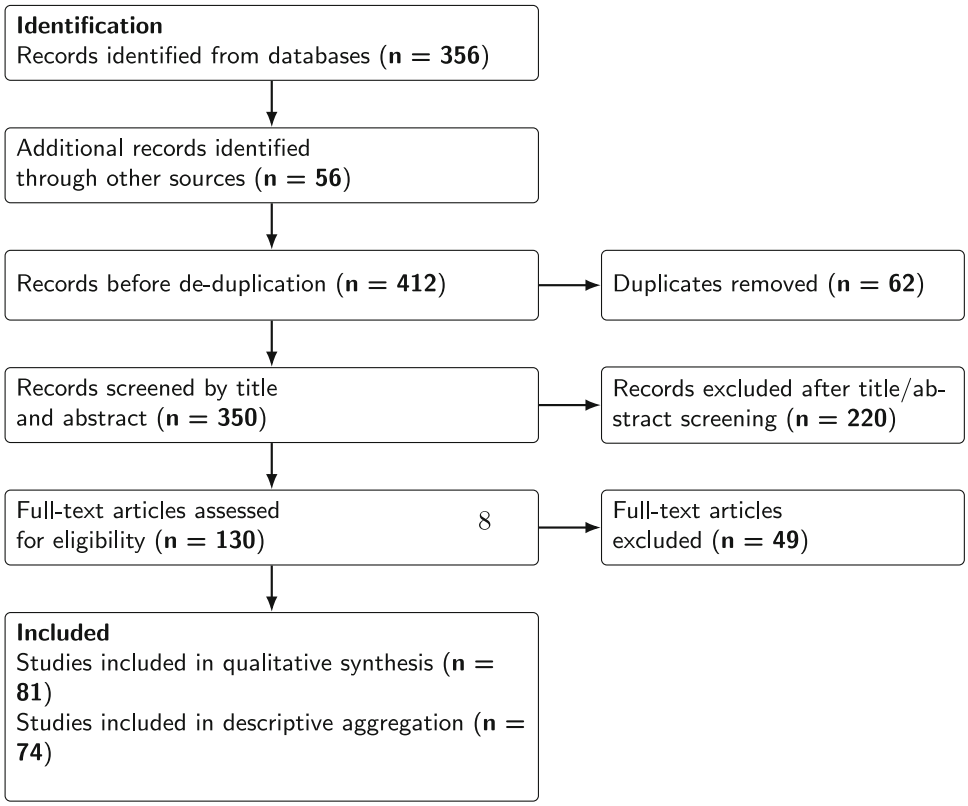


Fig. 2 PRISMA 2020 study selection flow diagram summarising literature identification, screening, eligibility assessment, and final study inclusion [9]

The search strategy was developed iteratively and refined through pilot searches to maximise retrieval of studies relevant to CNN-, Vision Transformer-, and hybrid CNN-ViT-based medical image classification while minimising irrelevant records.

Summary of the Related Literature

Deep learning has substantially advanced medical image classification through three dominant architectural paradigms: convolutional neural networks (CNNs), Vision Transformers (ViTs), and hybrid CNN–ViT models. Across the reviewed literature, these approaches demonstrate strong performance in applications including cancer detection, retinal disease analysis, neurological disorder diagnosis, and respiratory disease classification. However, as discussed in the "Discussion and Key Insights" section, performance remains strongly influenced by dataset characteristics, imaging modality, and evaluation protocol.

CNN-based models generally provide comparatively stable performance across diverse datasets due to their effectiveness in local feature extraction. Nevertheless, their limited ability to model long-range contextual dependencies has motivated the growing integration of attention mechanisms and transformer-based approaches. In contrast, ViT-based models demonstrate strong global contextual modelling capability, although their performance appears more sensitive to dataset scale and data quality. Hybrid CNN–ViT architectures aim to combine local and global feature representations, but often exhibit increased variability, architectural complexity, and dataset sensitivity.

A notable trend across recent studies is the growing adoption of attention mechanisms, particularly in CNN-based medical imaging models [13]. Attention modules improve feature refinement by emphasising diagnostically relevant regions, which is especially important in modalities characterised by subtle lesions and weak boundaries [14, 15]. Among these approaches, the Convolutional Block Attention Module (CBAM) has received considerable attention due to its lightweight integration of channel and spatial attention [16, 17]. Although CBAM improves feature representation, its pooling-based attention strategy may introduce information compression, limiting preservation of fine-grained local details.

The "Convolutional Neural Networks (CNNs)", "Vision Transformers (ViTs)", and "Hybrid CNN– ViT Models" sections provide a structured review of CNN-, ViT-, and hybrid-based medical image classification studies, focusing on recurring strengths, limitations, and unresolved challenges relevant to clinical robustness, interpretability, and generalisability.

Convolutional Neural Networks (CNNs)

CNNs remain the most established architecture in medical image classification due to their strong capability for hierarchical local feature extraction [18, 19]. Across the reviewed literature, CNN-based models achieve strong performance in applications including leukaemia, brain disease, lung disease, retinal imaging, breast cancer, liver disease, and skin lesion classification. Their effectiveness is particularly evident in imaging tasks where diagnostically relevant information is spatially localised.

Despite these strengths, several recurring limitations are observed. CNNs remain constrained in modelling long-range contextual dependencies, and their performance often depends on careful optimisation, sufficient annotated data, and transfer learning from natural-image datasets. In addition, many studies rely heavily on accuracy without comprehensive reporting of clinically relevant evaluation metrics, such as sensitivity (recall), specificity, precision, calibration, and uncertainty estimates, or external validation, limiting assessment of real-world generalisability.

The reviewed literature suggests that CNNs provide comparatively stable baseline performance across heterogeneous medical imaging settings, although their limitations continue to motivate the development of attention-enhanced and hybrid architectures.

Convolutional Block Attention Module (CBAM)

CBAM is a lightweight attention mechanism that sequentially applies channel and spatial attention to refine CNN feature maps by emphasising diagnostically relevant features while suppressing less informative responses [20]. In medical image classification, CBAM has been widely adopted to improve sensitivity to subtle lesions and anatomically relevant regions across modalities including X-ray, MRI, CT, ultrasound, and mammography [16, 21]. Its computational efficiency and ease of integration into existing CNN architectures have contributed to its widespread use in medical imaging applications.

Despite these advantages, several limitations constrain the clinical deployment of CBAM-enhanced models. First, the channel and spatial attention maps are generated using global average and max pooling, which may compress discriminative feature information and reduce sensitivity to subtle pathological structures, particularly small lesions, diffuse abnormalities, or poorly contrasted regions [20]. Second, because CBAM relies on attention weights learned from the training data, its effectiveness may decrease when image characteristics differ across institutions owing to variations in scanner manufacturers, acquisition protocols, reconstruction algorithms, or image quality. Such domain shifts may reduce model robustness and limit generalisability in routine clinical practice. Furthermore, although CBAM improves feature localisation, the generated attention maps should not be interpreted as definitive explanations of model decision-making, and additional explainability and calibration methods remain necessary to support clinician trust and safe deployment.

Consequently, recent research increasingly explores more adaptive attention and feature aggregation mechanisms that preserve fine-grained local information while improving robustness to heterogeneous clinical imaging conditions [13, 22]. These developments reflect the broader transition from lightweight attention modules towards more clinically robust feature modelling strategies for medical image analysis.

Vision Transformers (ViTs)

Vision Transformers (ViTs) model images using self-attention mechanisms rather than convolutional operations, enabling improved representation of long-range contextual relationships [23, 24]. The reviewed literature demonstrates strong performance across applications including respiratory disease diagnosis, retinal imaging, fracture classification, and skin lesion analysis [24–27].

Despite these advantages, ViTs remain strongly dependent on dataset scale and training quality. Their patch-based representation strategy may also reduce sensitivity to subtle local abnormalities, particularly in data-constrained medical imaging settings. In addition, several studies rely on natural-image pretraining, which may limit generalisability across institutions and imaging modalities.

ViTs represent a promising direction for modelling global contextual dependencies in medical imaging, although their clinical robustness remains closely linked to data availability, optimisation strategy, and preservation of local detail.

Hybrid CNN–ViT Models

Hybrid CNN–ViT architectures combine the locality and translation-related inductive biases of convolutional networks with the long-range contextual modelling of transformer-based attention. However, the reviewed literature does not employ a single form of hybridisation; rather, the models differ in where and how CNN and transformer representations are combined [28].

A common approach is *sequential or serial fusion*, in which a CNN first extracts local feature maps that are subsequently converted into tokens and processed by a transformer [29]. This design reduces the burden on the transformer to learn low-level visual structure and can be advantageous when medical datasets are limited. Nevertheless, information lost during convolution, downsampling, or tokenisation may not be recovered by the transformer, and the sequential pipeline can increase computational depth and inference latency.

A second strategy is *parallel or intermediate feature fusion*, in which CNN and transformer branches process the same image, or related representations, before their feature vectors are concatenated, added, gated, or combined through attention [30, 31]. Parallel fusion can retain complementary local and global information more directly than serial fusion, but it requires careful alignment of feature dimensions and scales. Simple concatenation may also preserve redundant or conflicting features, whereas attention- or gating-based fusion introduces additional parameters and optimisation complexity.

A less common approach is *late or decision-level fusion*, in which independently generated CNN and transformer predictions are combined through averaging, voting, weighting, or a secondary classifier. This strategy allows each branch to retain specialised representations and may simplify branch-level interpretation. However, because interaction occurs only after independent predictions have been produced, late fusion may not fully exploit relationships between local and global features, and its performance may depend strongly on the calibration and relative reliability of the individual branches [28].

From a clinical translation perspective, these fusion strategies should not be regarded as interchangeable, and no single strategy can be considered universally preferable. Sequential fusion may provide a comparatively straightforward and efficient integration of local and global representations, but information loss at the CNN–ViT interface can create an information bottleneck. Parallel or intermediate fusion may preserve complementary local and global representations more effectively, although the additional branches and feature-alignment operations increase memory, computational, and implementation requirements. Late fusion offers greater architectural modularity and may facilitate independent evaluation of component models. Still, limited cross-branch interaction and the need to combine potentially discordant predictions make its clinical reliability dependent on appropriate calibration and branch-level performance. Attention-guided and multi-scale fusion approaches attempt to address some of these limitations by adaptively weighting representations across branches and resolution levels [30, 31], although their additional complexity may complicate training, reproducibility, inference-time evaluation, and deployment. Consequently, the clinical suitability of a hybrid fusion strategy should be evaluated not only in terms of classification performance, but also with respect to calibration, computational requirements, robustness under domain shift, interpretability, and compatibility with the intended clinical workflow.

Accordingly, the descriptive variability observed among hybrid CNN–ViT studies may reflect not only dataset and modality differences but also substantial heterogeneity in fusion location, fusion operator, feature dimensionality, and training strategy. The findings of this review therefore do not support treating hybridisation as a uniformly superior design choice. Its value appears to depend on whether the selected fusion mechanism is appropriately matched to the imaging modality, dataset scale, target abnormality, and computational requirements of the intended clinical setting.

Hybrid CNN–ViT models represent a promising but evolving design space. Future progress will likely depend on improving efficiency, adaptive feature fusion, and robustness under heterogeneous clinical conditions.

Reporting of Results in Existing Literature

Performance metrics reported in the included studies were extracted as reported. Because accuracy, precision, recall, F_1-score, and specificity are conventionally expressed either as proportions (0–1) or percentages (0–100%), all percentage values were converted to proportions by dividing by 100, whereas metrics already reported on the 0–1 scale were retained unchanged. No further mathematical normalisation or rescaling of individual performance metrics was performed. Metrics were analysed separately, and only studies reporting a given metric were included in the corresponding descriptive aggregation. Performance metrics that were not reported by the original studies were recorded as "NA" and were excluded from metric-specific descriptive aggregation. Detailed study-level results are provided in Supplementary Tables 12–14 to support transparency and reproducibility. Collectively, these tables show that CNN studies constitute the largest and most modality-diverse evidence base, whereas ViT and hybrid CNN–ViT studies are smaller and more modality-concentrated. They also reveal variation in dataset scale, reported metrics, and validation practices across model families, providing important context for the descriptive cross-study comparisons.

Reported Results of CNN-Based Models

CNN-based models generally demonstrate strong and comparatively stable performance across multiple medical imaging tasks. Several studies report near-perfect accuracy and recall values, particularly in structured modalities such as OCT and microscopic imaging. Reported performance spans approximately 0.63–1.00 for accuracy and 0.70–1.00 for recall, reflecting variability across datasets and imaging modalities. The descriptive aggregation presented in the "Discussion and Key Insights" section further indicates that CNN-based models achieved the highest weighted accuracy (0.934) and weighted recall (0.901) (Table 7), while exhibiting comparatively lower accuracy variability ($s = 0.042$; Table 8) and recall variability ($s = 0.076$; Table 7).

Despite these strong results, reporting practices remain inconsistent. Many studies primarily report accuracy without complementary clinically relevant metrics such as specificity or calibration, while external validation remains limited (12.3%) across the literature.

Reported Results of ViT-Based Models

ViT-based models demonstrate competitive performance across several medical image classification tasks, particularly in OCT and X-ray imaging, where global contextual modelling may provide advantages. Several studies report high accuracy and recall values; however, performance variability across datasets suggests increased sensitivity to dataset scale, preprocessing, and data quality.

Reported performance generally spans approximately 0.73–1.00 for accuracy and 0.76–1.00 for recall. The descriptive aggregation additionally shows competitive weighted accuracy (0.906) and weighted recall (0.893), although with moderately higher recall variability ($s = 0.081$) compared to CNN-based models. These findings, together with the recurring identification of dataset dependence in the qualitative synthesis (Table 3), suggest that ViT-based models may be more sensitive to dataset size and quality than CNN-based models. A possible explanation is that transformer architectures rely primarily on self-attention and possess weaker built-in locality inductive biases than convolutional networks, making them more dependent on sufficiently large and diverse training datasets to learn robust visual representations [32, 33].

Reported Results of Hybrid CNN–ViT Models

Hybrid CNN–ViT models demonstrate strong reported performance across diverse medical imaging tasks by combining local convolutional feature extraction with global attention-based modelling. Several studies report high accuracy and recall; however, performance variability across datasets and modalities remains substantial.

Reported performance spans approximately 0.61–1.00 for accuracy and 0.30–1.00 for recall, representing the widest performance range among the reviewed model families. The descriptive aggregation presented in the "Discussion and Key Insights" section further indicates a weighted accuracy of 0.814, a weighted recall of 0.698, and the highest recall variability ($s = 0.168$) among the three model families. These findings suggest that, although hybrid architectures can achieve strong performance under certain conditions, their effectiveness remains more sensitive to dataset characteristics, architectural design, and optimisation strategy than either CNN- or ViT-based models.

These findings further suggest that integrating convolutional and transformer components does not inherently guarantee superior performance. Instead, the benefits of hybrid CNN–ViT architectures appear to depend on the imaging modality, dataset characteristics, and architectural implementation, indicating that hybridisation alone is insufficient to ensure improved generalisation.

Cross-Study Descriptive Aggregation

The combined qualitative and descriptive aggregation reveals several consistent patterns across model families. CNN-based models generally show the most consistent descriptive performance across modalities and dataset scales, whereas ViT-based models show competitive performance but appear more sensitive to dataset size and quality. Hybrid architectures provide flexible local–global feature representation; however, they exhibit greater variability across heterogeneous datasets.

These recurring strengths and limitations are summarised in Table 3. CNN-based models achieve the highest weighted accuracy and recall with comparatively lower recall variability; ViT-based models demonstrate moderate dataset sensitivity; and hybrid models exhibit substantially higher variability across studies.

In addition, the review highlights persistent inconsistencies in evaluation practices. Many studies continue to rely

Table 3 Qualitative cross-study comparison of CNN, ViT, and hybrid models for medical image classification. Reported performance ranges are descriptive and complement the descriptive aggregation

Model family	Accuracy range	Recall range	Common strengths	Recurring limitations
CNN	0.63–1.00	0.70–1.00	Strong local feature extraction; comparatively stable performance under limited data conditions	Limited global contextual modelling; limited external validation
ViT	0.73–1.00	0.76–1.00	Effective global contextual modelling; competitive performance in structured modalities	Higher data dependence; sensitivity to dataset quality and scale
Hybrid CNN–ViT	0.61–1.00	0.30–1.00	Combines local and global feature representations; flexible architectural design	Higher architectural complexity; increased dataset sensitivity and variability

primarily on accuracy without sufficient reporting of clinically relevant metrics such as recall, specificity, calibration, and uncertainty estimation. Furthermore, although several studies reported near-perfect performance, only 10 of the 81 reviewed studies (12.3%) performed independent external validation. Consequently, many reported performance values, including several perfect or near-perfect results, should be interpreted with caution, as they may reflect dataset-specific optimisation rather than genuine clinical generalisability.

Such near-perfect results may also indicate overfitting, particularly where model development and evaluation are confined to a single-centre dataset, repeated optimisation is performed on the same data distribution, or independent external testing is absent. High internal test performance should therefore not be interpreted as evidence of clinical generalisability without validation on geographically, demographically, and institutionally distinct populations.

The synthesis suggests that robust clinical translation will require not only strong classification performance, but also improved generalisability, standardised evaluation practices, uncertainty-aware modelling, and stronger external validation across heterogeneous clinical environments.

Discussion and Key Insights

This section presents a dataset-aware descriptive aggregation combined with cross-study qualitative analysis. A formal meta-analysis was not feasible because most included studies did not report sufficient study-level variance information required for meta-analytic pooling. Furthermore, substantial heterogeneity in imaging modalities, datasets, evaluation protocols, and reporting standards limited the appropriateness of formal inferential synthesis. Instead, paper-level aggregation was adopted to preserve dataset-specific context while enabling structured comparison across model families. Accordingly, the findings should be interpreted as a structured descriptive aggregation rather than an inferential meta-analysis.

Research Trends Across Model Families

As illustrated in Fig. 3, the reviewed literature demonstrates a clear progression in medical image classification research. CNN-based models dominated earlier studies due to their strong local feature extraction capability and comparatively stable performance under limited data conditions. ViT-based

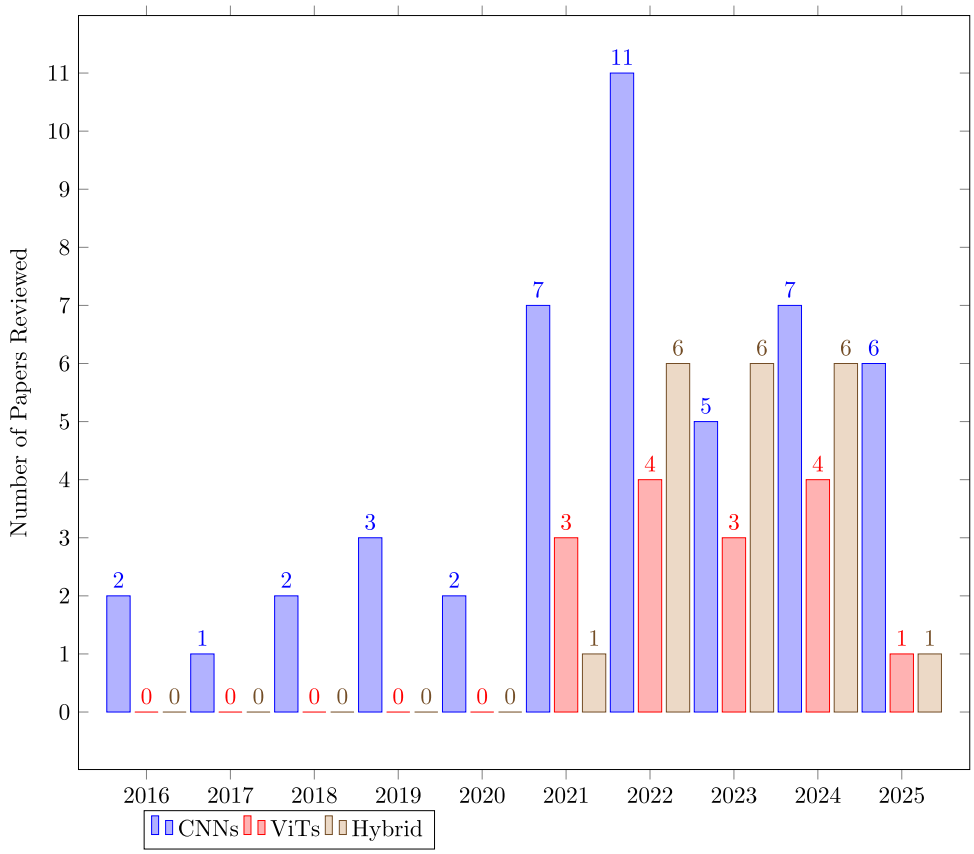


Fig. 3 Number of reviewed papers from 2016 to 2025

approaches emerged more recently, reflecting growing interest in modelling global contextual dependencies, whereas hybrid CNN–ViT architectures have increased rapidly since 2021, suggesting a shift towards combining local and global feature representations.

Descriptive Cross-Model Comparison

To enable structured comparison across heterogeneous studies, a paper-level aggregation strategy was adopted. When multiple datasets were reported within the same study, the dataset-size-weighted accuracy was computed as:

$$\hat{A}_{s,m} = \frac{\sum_{i=1}^{k} n_i A_i}{\sum_{i=1}^{k} n_i}, \tag{1}$$

where A_i denotes the reported accuracy for dataset i, n_i is the corresponding dataset size, and k represents the number of datasets or experiments reported within study s for imaging modality m. The resulting $\hat{A}_{s,m}$ represents the dataset-size-weighted study-level accuracy used for subsequent descriptive aggregation across studies.

Modality-level averages were subsequently computed using:

$$\bar{M}_m = \frac{1}{S_m} \sum_{s=1}^{S_m} \hat{M}_{s,m}, \tag{2}$$

where S_m denotes the number of studies evaluating modality m.

Overall accuracy was aggregated because it was consistently reported across nearly all included studies. Among the clinically relevant performance measures, recall (sensitivity) was selected for additional descriptive aggregation because it was the most consistently reported clinical metric after accuracy. Other clinically relevant metrics, including precision, F1-score, area under the receiver operating characteristic curve (AUC), calibration measures, and uncertainty estimates, were reported inconsistently enough to preclude meaningful descriptive aggregation without substantially reducing the number of eligible studies. Consequently, these metrics were synthesised qualitatively rather than quantitatively.

Globally weighted recall was further estimated as:

$$\bar{R}_w = \frac{\sum_{s=1}^{N} n_s R_s}{\sum_{s=1}^{N} n_s}, \tag{3}$$

where R_s and n_s denote the aggregated recall and total dataset size for study s, respectively.

Dataset scale sensitivity was assessed qualitatively based on the observed relationship between model performance and dataset size across studies.

Modality-Wise Descriptive Aggregation

Using the modality-level descriptive aggregation defined in Eq. 2, Tables 4, 5, and 6 summarise the aggregated performance across the three model families on a modality-wise basis. For each imaging modality, the tables report the mean performance together with the number of contributing studies (n) and the corresponding sample standard deviation (s). This analysis was based on the 74 studies included in the descriptive quantitative synthesis. CNN-based studies covered the widest range of imaging modalities. In contrast, ViT- and hybrid CNN–ViT studies were concentrated primarily on microscopic imaging, CT, X-ray, dermoscopy, OCT, and clinical photographic datasets.

Across all model families, OCT consistently demonstrated the highest mean accuracy (0.96–0.99) and mean recall (0.97–0.99), with relatively low standard deviations, indicating stable performance across studies. CT also achieved consistently high mean accuracy (0.93–0.95) and mean recall (0.90–0.92), although hybrid CNN–ViT models exhibited greater variability than CNN- and ViT-based models. In contrast, clinical photographic imaging and dermoscopy generally produced lower mean accuracies, particularly for ViT (0.77 and 0.91) and hybrid CNN–ViT models (0.83 and 0.73), while hybrid models also showed comparatively lower mean recall for ultrasound (0.82), X-ray (0.80), and clinical photographs (0.82).

From a cross-modality perspective, CNN-based models demonstrated the most consistent performance, with generally lower accuracy and recall standard deviations across the broadest range of imaging modalities. ViT-based models achieved excellent performance for OCT and X-ray imaging but exhibited greater variability in heterogeneous modalities such as dermoscopy and clinical photography. Hybrid CNN–ViT models showed the largest performance fluctuations, particularly for X-ray (accuracy $s = 0.138$, recall $s = 0.258$) and ultrasound (recall $s = 0.160$), despite achieving competitive performance for CT and OCT.

Although a formal correlation analysis was beyond the scope of this descriptive review, the included studies suggest that dataset size may have influenced reported performance across imaging modalities. Modalities such as OCT and CT, which were represented by multiple studies and frequently employed comparatively large, well-curated datasets, generally exhibited higher and more consistent mean accuracy and recall. Conversely, clinical photographic imaging, dermoscopy, and ultrasound often relied on smaller or more heterogeneous datasets and demonstrated greater performance variability. These observations suggest that differences in

Table 4 Modality-wise descriptive aggregation for the CNN models. Values are reported as mean performance; the number of contributing studies (n) is shown in parentheses

Modality Modality	Mean Accuracy	Accuracy Std. (s)	Mean Recall	Recall Std. (s)
Microscopic	0.96 (11)	0.035	0.95 (8)	0.054
Ultrasound	0.92 (4)	0.056	0.93 (3)	0.095
MRI	0.90 (13)	0.086	0.85 (7)	0.105
CT	0.95 (13)	0.041	0.92 (5)	0.061
X-ray	0.89 (13)	0.104	0.95 (8)	0.043
Dermoscopy	0.89 (5)	0.092	0.94 (11)	0.031
OCT	0.96 (6)	0.042	0.97 (4)	0.037
Clinical Photo	0.89 (2)	0.149	0.99 (1)	ND
Scintigraphy	0.98 (1)	ND	0.98 (1)	ND
Otoendoscopy	0.92 (1)	ND	0.00 (0)	ND

Table 6 Modality-wise descriptive aggregation for the hybrid CNN–ViT models. Values are reported as mean performance; the number of contributing studies (n) is shown in parentheses

Modality Modality	Mean Accuracy	Accuracy Std. (s)	Mean Recall	Recall Std. (s)
Microscopic	0.92 (22)	0.077	0.92 (16)	0.069
Ultrasound	0.88 (4)	0.074	0.82 (3)	0.160
MRI	0.00 (0)	NIL	0.00 (0)	NIL
CT	0.93 (12)	0.105	0.90 (9)	0.134
X-ray	0.86 (10)	0.138	0.80 (8)	0.258
Dermoscopy	0.73 (1)	ND	0.00 (0)	NIL
OCT	0.97 (4)	0.037	0.99 (3)	0.014
Clinical photo	0.83 (5)	0.095	0.82 (5)	0.112
Scintigraphy	0.00 (0)	NIL	0.00 (0)	NIL
Otoendoscopy	0.00 (0)	NIL	0.00 (0)	NIL

reported performance are likely influenced not only by model architecture but also by dataset scale and underlying data characteristics.

Several modalities, including scintigraphy, otoendoscopy, and dermoscopy in the hybrid model family, were represented by only one or no studies, resulting in undefined or unavailable standard deviations. Consequently, findings for these underrepresented modalities should be interpreted cautiously.

Weighted Global Performance Comparison

In this review, dataset sensitivity refers to the extent to which reported model performance varied with the size, quality, and characteristics of the training data, as interpreted from the descriptive aggregation, qualitative synthesis, and observed performance variability across the included studies. It should therefore be regarded as an interpretive rather than a formally estimated quantitative measure. The weighted global performance comparison (Table 7) was derived from all 74 studies for weighted accuracy. Weighted recall was computed only from studies reporting recall (28 CNN, 10 ViT, and 19 hybrid CNN–ViT studies). CNN-based models achieved the highest weighted accuracy (Eq. 1) and weighted recall (Eq. 3) with comparatively lower recall variability, suggesting greater robustness across heterogeneous datasets and imaging modalities. ViT-based models also demonstrated competitive aggregated performance, particularly in imaging settings where global contextual modelling may be advantageous. Hybrid CNN–ViT models exhibited substantially higher recall variability and lower weighted recall, indicating greater observed sensitivity to dataset characteristics, optimisation strategies, and dataset scale across the reviewed studies. Although several individual studies reported perfect or near-perfect performance (accuracy or recall = 1.00 or approaching 1.00), these findings were predominantly obtained using single-centre datasets without independent external validation. Consequently, such results should be interpreted cautiously, as they may reflect dataset-specific optimisation, limited dataset diversity, or potential overfitting rather than robust clinical generalisability.

The weighted analysis suggests that increased architectural complexity does not necessarily correspond to consistently improved performance across heterogeneous medical imaging settings.

Table 5 Modality-wise descriptive aggregation for the ViT models. Values are reported as mean performance; the number of contributing studies (n) is shown in parentheses

Modality Modality	Mean Accuracy	Accuracy Std. (s)	Mean Recall	Recall Std. (s)
Microscopic	0.87 (5)	0.059	0.92 (2)	0.057
Ultrasound	0.89 (3)	0.060	0.91 (2)	0.089
MRI	0.00 (0)	NIL	0.00 (0)	NIL
CT	0.94 (2)	0.092	0.92 (2)	0.120
X-ray	0.99 (4)	0.013	0.93 (5)	0.101
Dermoscopy	0.91 (6)	0.068	0.82 (1)	ND
OCT	0.99 (2)	0.021	0.99 (2)	0.021
Clinical photo	0.77 (2)	0.050	0.00 (0)	NIL
Scintigraphy	0.00 (0)	NIL	0.00 (0)	NIL
Otoendoscopy	0.00 (0)	NIL	0.00 (0)	NIL

Descriptive Variability Analysis

The descriptive variability analysis for accuracy (Table 8) included all 74 studies reporting comparable accuracy values, whereas the recall analysis (Table 9) included the 57 studies that reported recall. The descriptive summary statistics

Table 7 Weighted performance comparison across model families. Weighted accuracy was obtained using the modality-level descriptive aggregation (Eq. 1), whereas weighted recall was computed using (Eq. 3)

Model	Studies	Weighted accuracy	Weighted recall	Recall std. (s)	Recall range	Interpretive dataset sensitivity
CNN	42	0.934	0.901	0.076	0.70–1.00	Low
ViT	12	0.906	0.893	0.081	0.76–1.00	Medium
Hybrid	20	0.814	0.698	0.168	0.30–1.00	High

presented in Tables 8 and 9 were computed using Eqs. 4–6. Specifically, the mean performance was estimated as:

$$\bar{x} = \frac{1}{N}\sum_{j=1}^{N} x_j, \tag{4}$$

where x_j denotes the aggregated performance value for paper j, and N is the number of eligible papers.

The sample standard deviation was computed as:

$$s = \sqrt{\frac{1}{N-1}\sum_{j=1}^{N}(x_j - \bar{x})^2}, \tag{5}$$

The interquartile range (IQR) was computed as:

$$IQR = Q_3 - Q_1. \tag{6}$$

The standard deviations (Eq. 5) and interquartile ranges (Eq. 6) reported in Tables 8 and 9 describe the dispersion of the aggregated paper-level performance values across the included studies. These statistics are presented solely as descriptive measures of variability and should not be interpreted as estimates of between-study statistical heterogeneity, such as Cochran's Q, τ^2, or I^2, which require study-level variance information that was unavailable for most included studies. Accordingly, Cochran's Q, τ^2, and I^2 were not calculated, as their application without the required study-level variance information and sufficiently comparable effect estimates could provide a misleading representation of between-study heterogeneity. Formal significance tests between model families were likewise not performed because the studies differed substantially in datasets, imaging modalities, disease targets, sample sizes, and evaluation protocols; model-family comparisons are therefore intentionally descriptive rather than inferential. The observed variability reflects differences in datasets, imaging modalities, disease categories, experimental protocols, and model implementations across the reviewed studies.

Sensitivity Analysis

To evaluate the robustness of the descriptive quantitative synthesis, a simple sensitivity analysis was performed by comparing the primary dataset-size-weighted aggregation with an alternative equal-weighted aggregation, in which each study contributed equally regardless of dataset size. This comparison assessed whether the principal findings were sensitive to the weighting strategy adopted for the descriptive synthesis.

Table 10 compares the weighted and equal-weighted aggregated accuracy and recall across the three model families. Overall, the differences between the two aggregation approaches were minimal, with absolute differences ranging from 0.006 to 0.007 for both accuracy and recall. CNN-based models remained the highest-performing model family under both weighting strategies, followed by ViT models and hybrid CNN–ViT models. The relative ranking of the three model families was therefore unchanged, indicating that the principal findings of the descriptive quantitative synthesis were robust to the choice of aggregation method.

The equal-weighted aggregation produced slightly lower accuracy and recall for CNN and ViT models, whereas hybrid CNN–ViT models exhibited marginally higher aggregated performance. However, these numerical differences were negligible and did not alter the overall interpretation of the comparative performance of the three model families.

The sensitivity analysis specifically evaluated the influence of the aggregation weighting strategy and does not account for other potential sources of bias, including study

Table 8 Descriptive variability summary for paper-level accuracy computed using Eqs. 4–6

Model	n	Mean	Median	Std. (s)	IQR
CNN	42	0.928	0.952	0.042	0.078
ViT	12	0.899	0.907	0.054	0.112
Hybrid	20	0.821	0.836	0.071	0.118

Table 9 Descriptive variability summary for paper-level recall computed using Eqs. 4–6

Model	n	Mean	Median	Std. (s)	IQR
CNN	28	0.895	0.918	0.076	0.097
ViT	10	0.887	0.900	0.081	0.116
Hybrid	19	0.705	0.731	0.168	0.152

Table 10 Sensitivity analysis comparing dataset-size-weighted and equal-weighted descriptive aggregations

Model Family	Weighted Accuracy	Equal-weighted Accuracy	Δ Accuracy	Weighted Recall	Equal-weighted Recall	Δ Recall
CNN	0.934	0.928	−0.006	0.901	0.895	−0.006
ViT	0.906	0.899	−0.007	0.893	0.887	−0.006
Hybrid						
CNN–ViT	0.814	0.821	+0.007	0.698	0.705	+0.007

quality, publication bias, selective outcome reporting, dataset composition, or validation design. These factors could not be formally assessed because sufficiently standardised study-level information was unavailable. Future evidence syntheses could investigate quality-weighted, modality-specific, or variance-based aggregation and publication bias where sufficient data are available.

Current Practices and Clinical Translation

Evaluation Practices and Limitations

The reviewed literature demonstrates continued reliance on accuracy despite its limitations in imbalanced medical datasets. High overall accuracy may conceal poor detection of clinically important minority or disease-positive classes. In such settings, sensitivity (recall) reflects detection of positive cases, specificity reflects performance on negative cases, precision the reliability of positive predictions, and the F1-score the balance between precision and recall. Clinically relevant metrics such as recall, specificity, and precision remain inconsistently reported, as shown in Fig. 4, limiting reliable cross-study comparison and clinical interpretability. Future studies should therefore report class distributions alongside class-sensitive metrics and, where appropriate, per-class performance, calibration, uncertainty, and confusion matrices.

Dataset Accessibility and Reproducibility

Approximately 70% of studies utilised publicly available datasets, supporting benchmarking and reproducibility. Nevertheless, substantial variability remains in dataset preprocessing, annotation quality, and evaluation protocols, limiting comparability and external validation across studies. Reproducibility also depends on transparent reporting of the complete modelling pipeline, as differences in preprocessing, augmentation, data partitioning, hyperparameter selection, model configuration, and evaluation procedures can affect reported performance even when the same dataset is used. Studies should therefore report these procedures in sufficient detail and, where possible, provide code, model configurations, random seeds, and clearly defined dataset splits. Importantly, reproducibility should be distinguished from external generalisability: a model may be reproducible on its original dataset without maintaining performance across independent institutions or patient populations.

Implications for Clinical Translation

The reviewed literature demonstrates that strong retrospective classification performance alone is insufficient to establish clinical readiness. Only 10 of the 81 reviewed studies (12.3%) reported independent external validation, indicating that the vast majority of published models were evaluated solely on internal datasets. Consequently, the comparative findings of this review should be interpreted as evidence of reported model performance rather than direct evidence of clinical effectiveness. Without validation across institutions, patient populations, or imaging acquisition settings, reported performance may overestimate real-world clinical generalisability. Translation into practice therefore requires evaluation of model robustness under variations in disease prevalence, imaging equipment, acquisition protocols, and clinical workflows [34, 35].

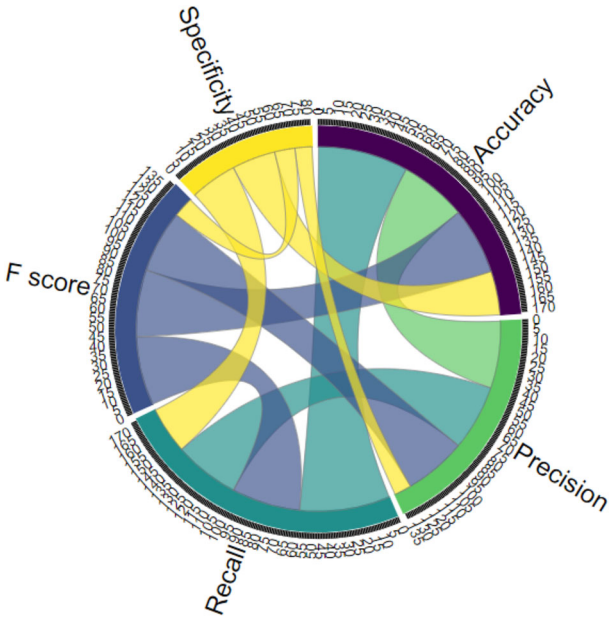


Fig. 4 Frequency of evaluation metrics used across studies. An interactive version of this Figure can be found here: https://carlosfmorenog.github.io/chord/

The intended clinical role of an algorithm should also be defined before deployment. A model used for worklist prioritisation, for example, may require very low latency and high sensitivity, whereas a second-reader system used during reporting may tolerate longer processing times but require more detailed explanations and calibrated confidence estimates. Similarly, tools intended for point-of-care ultrasound, mobile screening, or resource-constrained settings may face stricter limitations on memory, energy consumption, connectivity, and hardware availability than systems operating on central hospital servers. These differences indicate that clinical utility cannot be inferred from architecture family alone and should instead be evaluated in relation to the intended use, operating environment, and consequences of model error.

Computational Efficiency and Inference Time

Computational efficiency represents an important but inconsistently reported aspect of clinical deployment. The reviewed studies generally emphasised predictive performance, while inference latency, throughput, memory consumption, model size, and energy use were rarely reported in a sufficiently standardised form for cross-study comparison. Consequently, this review cannot determine whether CNN-, ViT-, or hybrid CNN–ViT models are consistently more efficient in clinical operation. Although lightweight CNN and transformer variants have been proposed within the reviewed literature [25, 36], reported efficiency may depend as strongly on image resolution, preprocessing, hardware, software implementation, batch size, and deployment setting as on the underlying architecture.

For clinically meaningful evaluation, inference time should include the complete processing pathway rather than model execution alone. Relevant components may include image retrieval, preprocessing, data transfer, model inference, post-processing, result communication, and integration with the clinical information system. Reporting only accelerator-level inference time may therefore underestimate the delay experienced by clinicians. Future studies should report, where applicable, median and upper-percentile latency, throughput, model size, parameter count, memory demand, computational operations, hardware specification, batch size, and image dimensions. Measurements using a batch size of one may be particularly relevant to real-time or patient-level clinical use.

Model compression methods, including quantisation, pruning, knowledge distillation, and low-rank approximation, may improve deployment feasibility by reducing memory and computational requirements. However, efficiency gains should not be assumed to preserve clinical performance. Compressed models should be reassessed for sensitivity, specificity, calibration, subgroup performance, robustness under acquisition variation, and failure behaviour. The appropriate objective is therefore not simply the smallest or fastest model, but a model that achieves an acceptable balance among predictive performance, latency, resource consumption, reliability, and maintainability in its intended clinical environment.

Clinical Workflow Integration of Medical Image Classification

Medical image classification systems must operate within established clinical processes rather than as isolated prediction models. Table 11 illustrates the potential roles of CNN-, ViT-, and hybrid-based models across different stages of the clinical workflow, ranging from image-quality assessment and diagnostic support to disease monitoring, treatment planning, and integrated clinical decision support. However, the model assignments presented in Table 11 are intended to be illustrative rather than prescriptive, because suitability depends on the clinical task, available evidence, infrastructure, and intended level of human oversight.

Workflow Design, Human Factors, and Prospective Evaluation

Effective integration requires specification of where the model is activated, which data it receives, how its output is presented, who is responsible for reviewing the result, and what action should follow an uncertain or erroneous

Table 11 Role of AI models across clinical workflow stages

Clinical stage	AI function	Suitable model type	Clinical impact
Image acquisition	Quality assessment, preprocessing	CNN	Improved image reliability
Diagnosis	Lesion detection and classification	CNN/Hybrid	Early and accurate diagnosis
Disease assessment	Extent and severity analysis	ViT/Hybrid	Better clinical interpretation
Treatment planning	Subtype classification, prognosis	ViT/Hybrid	Personalised therapy decisions
Monitoring	Longitudinal progression tracking	Hybrid	Treatment effectiveness evaluation
Clinical decision Support	Integrated diagnosis and treatment	Hybrid + multimodal AI	End-to-end workflow support

prediction. In radiology, pathology, and ophthalmology, this may require integration with picture archiving and communication systems, reporting workstations, laboratory information systems, or electronic health records. Poorly positioned alerts or outputs that require clinicians to leave their normal workspace may increase cognitive burden and reduce adoption even when technical performance is strong.

Human–AI interaction should therefore be evaluated as part of system performance. Relevant outcomes include changes in reporting time, workload, diagnostic agreement, override behaviour, alert fatigue, and the frequency with which clinicians appropriately accept or reject model recommendations. Early deployment may initially be conducted in a silent mode, in which predictions are generated without influencing care, before progressing to prospective evaluation with clinician interaction. The DECIDE-AI guideline provides recommendations for reporting early-stage clinical evaluations of AI-based decision-support systems, including assessment of actual clinical performance, safety, workflow interaction, and human factors [37]. Subsequent comparative or interventional studies should follow relevant AI-specific protocol and trial-reporting guidance, including SPIRIT-AI and CONSORT-AI [38, 39].

Deployment should also include mechanisms for identifying out-of-distribution inputs, communicating uncertainty, recording model and user actions, and escalating cases that require specialist review. AI outputs should support rather than obscure clinical accountability. The level of automation should therefore reflect the severity and reversibility of potential errors, with higher-risk decisions generally requiring stronger clinical oversight and prospective evidence.

Regulatory Approval and Lifecycle Management

Medical image classification software intended to support diagnosis, triage, prognosis, or treatment decisions may be regulated as medical-device software, depending on its intended use and jurisdiction. In the United States, the applicable Food and Drug Administration (FDA) marketing pathway may include premarket notification (510(k)), De Novo classification, or premarket approval, depending on device classification, novelty, risk, and the availability of an appropriate predicate [40–42]. Regulatory clearance or approval should not be treated as a one-time endpoint; it forms part of a broader product lifecycle that includes quality management, version control, post-market monitoring, cybersecurity, and management of software changes.

Recent FDA guidance for AI-enabled device software functions emphasises a total product lifecycle approach, including documentation of data management, model development, validation, risk management, transparency, and post-market performance [43]. Good Machine Learning Practice principles similarly emphasise representative datasets, independence between training and test data, evaluation of the human–AI team, clear information for users, and monitoring of deployed models [44]. For models designed to change after deployment, a predetermined change control plan may provide a structured description of anticipated modifications and the procedures used to evaluate them [45].

Regulatory evidence should therefore correspond to the model's intended clinical claim. Retrospective performance on a public benchmark is unlikely to be sufficient for a system intended to affect patient management. Evidence may need to include analytical validation, external and prospective clinical validation, usability testing, subgroup analysis, software verification, cybersecurity assessment, and a plan for detecting performance degradation. Models should also retain traceable versioning so that predictions can be linked to the software, model weights, decision threshold, and data-processing pipeline used at the time of inference.

Privacy, Cybersecurity, and Federated Learning

Medical imaging datasets may contain directly identifiable information as well as biological features that remain sensitive after removal of conventional identifiers. Clinical deployment therefore requires data minimisation, role-based access control, encryption, auditability, secure data transfer, and governance arrangements covering data retention, secondary use, and accountability. Privacy and security should be considered throughout the system lifecycle rather than only during dataset collection. This is particularly important when models interact with hospital networks, cloud services, imaging archives, or electronic health records. FDA cybersecurity guidance similarly treats cybersecurity as a lifecycle concern extending across device design, deployment, maintenance, and post-market monitoring [46]. Broader ethical guidance also emphasises autonomy, transparency, accountability, inclusiveness, and protection of patient rights [47].

Federated learning offers one potential approach for multi-institutional model development by allowing participating organisations to train a shared model without centrally pooling their raw patient data. Studies have demonstrated the feasibility of federated learning across multiple healthcare institutions and imaging datasets [48, 49]. This approach may facilitate access to more diverse populations and acquisition protocols while reducing the need to transfer source images outside the institutions that hold them.

Nevertheless, federated learning should not be described as providing complete privacy by default. Model updates may still disclose information, and participating institutions may differ substantially in disease prevalence, imaging

protocols, annotation practices, computational capacity, and data quality. These non-independent and non-identically distributed data can reduce convergence and produce unequal performance between sites. Federated systems also introduce communication costs, governance requirements, participant-authentication risks, and questions concerning ownership and responsibility for the resulting model. Where appropriate, federated learning may therefore need to be combined with secure aggregation, differential privacy, encryption, access controls, and formal agreements defining participation, auditing, update management, and incident response.

Figure 5 summarises this clinical translation pathway, illustrating how data quality, model development, validation, explainability, regulation, workflow integration, and post-deployment monitoring collectively influence successful clinical adoption. Clinical translation should therefore be evaluated as a lifecycle process rather than inferred solely from retrospective classification performance.

Future Research Directions

The quantitative and qualitative synthesis presented in the "Discussion and Key Insights" section identified recurring challenges related to dataset dependency, modality-specific variability, limited external validation, inconsistent reporting, and clinical deployment. The future research priorities presented below are derived directly from these findings and address the principal methodological and translational limitations identified across the reviewed studies.

Data-Efficient Learning Under Limited Annotation

The descriptive aggregation showed that ViT- and hybrid-based architectures generally exhibited greater performance variability and appeared more sensitive to dataset size and quality than CNN-based models. These findings suggest that limited availability of large, well-annotated medical datasets remains an important constraint for reliable model development. Future research should therefore prioritise data-efficient learning strategies, including self-supervised learning, self-distillation, few-shot learning, and domain-specific pretraining, to reduce dependence on extensive manual annotation while improving robustness and generalisability.

Dataset Bias and Representative Data

The descriptive aggregation identified substantial dataset dependency and limited external validation across the reviewed

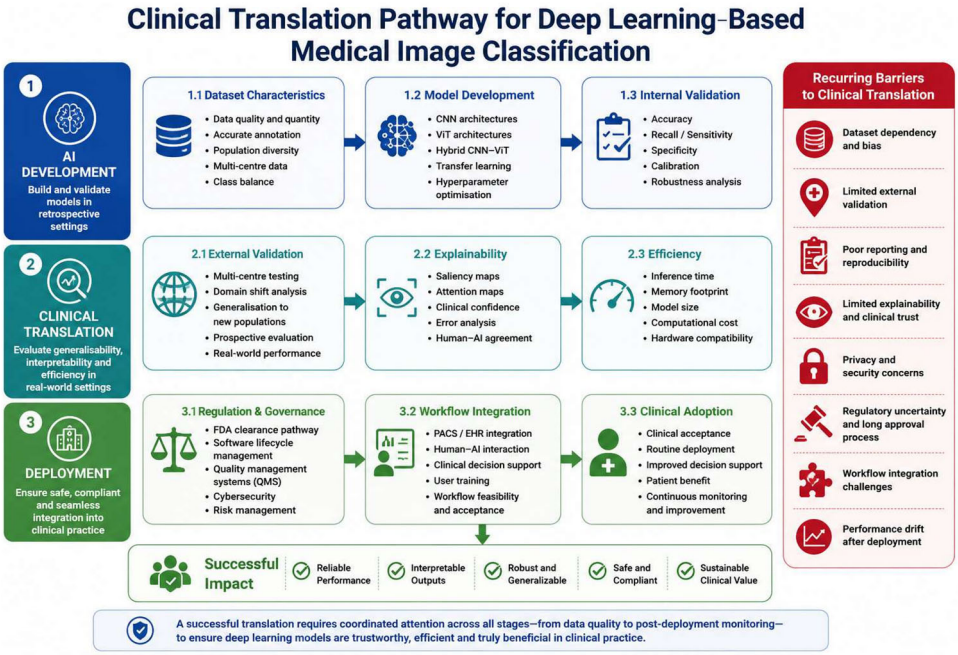


Fig. 5 Clinical translation pathway for deep learning-based medical image classification systems. The framework illustrates the progression from AI development through clinical translation and deployment to clinical adoption, together with the recurring barriers that may limit successful implementation

studies, suggesting that reported performance may be influenced by the characteristics of the underlying training populations rather than architecture alone. Demographic imbalances related to age, sex, ethnicity, disease prevalence, and imaging-site representation, as well as differences in image acquisition protocols, scanner manufacturers, annotation practices, and institution-specific workflows, may reduce model generalisability across diverse clinical settings [34, 35]. Such under-representation can result in systematic differences in model performance across patient subgroups, even when overall accuracy appears high. This is particularly important for clinical deployment because aggregate performance metrics may conceal reduced sensitivity or increased error rates in under-represented populations. Future evaluations should therefore report demographic composition where available and assess subgroup performance across relevant characteristics such as age, sex, ethnicity, and other clinically meaningful population factors.

The rapid development of AI models during the COVID-19 pandemic further illustrated these challenges, as many studies relied on relatively small, single-centre, or publicly aggregated datasets that showed limited generalisability across different populations and acquisition settings [50, 51]. Future studies should therefore report dataset demographics and acquisition characteristics more comprehensively, evaluate performance across clinically relevant patient subgroups, and perform multi-centre external validation to improve fairness, robustness, and generalisability.

Adaptive and Multi-Scale Feature Representation

The modality-wise descriptive aggregation indicated greater performance variability for dermoscopic and clinical photographic imaging than for CT, MRI, and OCT, suggesting that existing architectures do not consistently capture both fine-grained local features and broader contextual information across heterogeneous imaging conditions. Future research should therefore investigate adaptive multi-scale feature learning, including hierarchical attention, deformable feature extraction, and context-aware feature fusion, to improve robustness and generalisability across diverse imaging modalities.

Interpretable and Clinically Meaningful Explanations

The qualitative synthesis indicated that explainability analyses were reported less consistently than conventional performance metrics, limiting assessment of whether model predictions are clinically trustworthy. Future research should therefore incorporate explainability as a routine component of model evaluation. Common post-hoc techniques include Gradient-weighted Class Activation Mapping (Grad-CAM), which produces class-specific localisation maps from convolutional features [52]; Local Interpretable Model-Agnostic Explanations (LIME), which approximates individual predictions using a locally interpretable surrogate model [53]; and Shapley Additive exPlanations (SHAP), which assigns feature contributions using a game-theoretic framework [54]. Attention visualisation is also frequently used for transformer and hybrid models to indicate image regions or patches receiving greater model attention.

Although these methods may provide useful qualitative insight, they should not be interpreted as direct evidence of causal reasoning or clinical validity. Grad-CAM maps are typically coarse and may vary with model layer or implementation; LIME can be sensitive to perturbation and segmentation choices; SHAP may be computationally expensive and dependent on the selected background distribution; and attention weights do not necessarily correspond to feature importance or faithfully explain model predictions [55]. Future studies should therefore evaluate explanation stability, faithfulness, anatomical plausibility, and agreement with clinical expertise, preferably across external datasets and clinically relevant patient subgroups.

Transfer Learning and Medical Foundation Models

Across the reviewed studies, pretraining strategies varied, including conventional transfer learning from natural-image datasets such as ImageNet and, in a smaller emerging body of work, domain-specific or medical-image pretraining. The available evidence was not sufficiently standardised for formal comparison because studies differed in architecture, imaging modality, dataset size, fine-tuning protocol, and evaluation design. Nevertheless, the qualitative evidence suggests that the effectiveness of ImageNet pretraining is task-dependent and should not be assumed to transfer uniformly to medical imaging applications [56].

Medical-domain pretraining and self-supervised learning offer a promising alternative because representations can be learned from image distributions more closely aligned with clinical anatomy, pathology, and acquisition characteristics [57]. However, the reviewed evidence does not establish their consistent superiority over ImageNet pretraining. Direct comparisons using matched architectures, datasets, fine-tuning protocols, and external validation are therefore needed to determine when medical-domain pretraining provides clinically meaningful advantages.

More recently, medical foundation models have emerged as a promising direction for learning generalisable representations across multiple imaging modalities and downstream

applications [58]. However, evidence regarding their superiority over conventional transfer learning remains limited, and further comparative evaluation under consistent external validation protocols is required before their advantages can be established across diverse clinical settings.

Multimodal Artificial Intelligence

The qualitative synthesis showed that most included studies focused exclusively on image-based classification, with comparatively limited integration of complementary clinical information. This suggests that current performance may not fully exploit the broader patient context available during routine clinical decision-making. Future research should therefore investigate multimodal AI frameworks that combine medical images with clinical variables (e.g., demographics, laboratory measurements, and electronic health records), molecular and omics data, and other relevant patient information to support more comprehensive clinical assessment.

Building upon multimodal learning, recent advances in large language models (LLMs) further provide opportunities to integrate imaging findings with clinical documentation, radiology reports, and decision-support systems. However, the clinical utility of such multimodal systems remains to be established through rigorous external validation, transparent evaluation, and assessment of their effects on clinical workflow and patient outcomes.

Uncertainty-Aware and Trustworthy AI

The descriptive aggregation showed that conventional performance metrics, such as accuracy, precision, recall, and F1-score, were reported considerably more frequently than uncertainty estimation, confidence calibration, and external validation. These findings suggest that predictive performance alone may be insufficient to establish clinical trustworthiness under heterogeneous clinical conditions. Future research should therefore incorporate uncertainty-aware learning, confidence calibration, and robust evaluation strategies to better characterise model reliability, particularly under domain shift and out-of-distribution settings.

Clinically Robust and Generalisable Systems

The descriptive aggregation revealed considerable performance variability across model families, imaging modalities, and datasets, while the methodological assessment identified limited external validation and frequent reliance on single-centre datasets. These findings indicate that model performance may vary under different patient populations, imaging devices, and acquisition protocols. Future research should therefore prioritise multi-centre external validation, systematic evaluation under domain shift, and standardised benchmarking using clinically representative datasets and harmonised evaluation protocols to improve the robustness and generalisability of medical image classification systems.

Clinical Workflow Integration and Deployment

The methodological assessment identified limited reporting of deployment-related outcomes despite generally strong retrospective classification performance. First, studies should evaluate computational feasibility by reporting end-to-end latency, memory demand, hardware requirements, and throughput under conditions representative of the intended clinical setting. Second, externally validated models should progress to prospective and human-factors evaluation to determine their effects on workflow, clinical decisions, workload, and patient safety. Third, development should incorporate regulatory, privacy, cybersecurity, and post-deployment monitoring requirements from an early stage rather than treating them as considerations that arise only after model training.

Multi-institutional and privacy-preserving approaches, including federated learning, may support broader validation without routine centralisation of patient data. Still, they require explicit assessment of site heterogeneity, privacy leakage, governance, and communication burden. Future deployment studies should also define the intended user, degree of automation, failure-response pathway, and procedures for detecting domain shift and performance degradation. These priorities are reflected in Fig. 5, which presents clinical adoption as the outcome of coordinated progress across data, modelling, validation, explainability, regulation, and workflow integration.

Prioritisation of Future Research

Taken together, the findings of this review indicate that future research should prioritise the development of more robust, data-efficient, and clinically adaptable medical image classification frameworks. In particular, the complementary strengths of CNNs in local feature extraction and ViTs in global contextual modelling, together with the greater performance variability observed among existing hybrid CNN–ViT models, motivate further investigation of adaptive and computationally efficient fusion strategies that can exploit local–global representations without introducing excessive architectural complexity or dataset sensitivity. Such methodological development should be supported by data-efficient and domain-specific pretraining, interpretable and uncertainty-aware prediction, and evaluation across heterogeneous imaging modalities and patient

populations. External and prospective validation, representative datasets, and standardised reporting should provide the evidence required to determine whether these methodological advances translate into robust, generalisable, and clinically meaningful performance.

Limitations of the Study

This review has some limitations. First, a formal statistical meta-analysis was not feasible because most included studies did not report the study-level variance statistics required for variance-based pooling. Consequently, the quantitative findings should be interpreted as descriptive comparative summaries rather than pooled effect estimates. Second, substantial heterogeneity in imaging modalities, datasets, preprocessing methods, evaluation protocols, and validation strategies limited direct cross-study comparability; therefore, the reported comparisons should be interpreted as descriptive comparative trends rather than direct head-to-head evaluations. Third, because the review relied on published studies, publication bias cannot be excluded, and unpublished or negative findings (the "file-drawer" problem) were not represented in the descriptive synthesis. Finally, restricting the review to English-language publications may have excluded relevant studies. These limitations should be considered when interpreting the findings and identifying future research directions.

Additional limitations should also be recognised. The review-specific methodological quality assessment was designed to accommodate the heterogeneous study designs included in this review and does not constitute a validated diagnostic-study risk-of-bias instrument. The restriction to Q1 journals and A/A* conferences was adopted as a pragmatic quality-oriented eligibility criterion but is not a standard requirement of systematic-review methodology and may have excluded relevant evidence from other publication venues. Furthermore, formal inter-rater reliability statistics were not calculated during study screening. Finally, the descriptive sensitivity analysis evaluates the influence of aggregation weighting but does not quantify the effects of study quality, publication bias, or selective outcome reporting on the observed performance patterns.

Conclusion

This review presents a structured qualitative synthesis complemented by a dataset-aware descriptive quantitative aggregation of CNN, ViT, and hybrid CNN–ViT-based medical image classification studies published between 2016 and 2025. By integrating descriptive aggregation with cross-study analysis, the review provides a clinically grounded comparison of performance trends, variability, dataset dependency, and methodological characteristics across the three major model families.

The findings indicate that CNN-based models generally demonstrated the most consistent performance across heterogeneous imaging settings, whereas ViT- and hybrid CNN–ViT architectures achieved competitive results in several modalities but appeared more sensitive to dataset scale, optimisation strategy, and data quality. The synthesis also revealed substantial variability in reported performance and persistent inconsistencies in evaluation practices, particularly the limited reporting of clinically relevant metrics such as recall, specificity, calibration, uncertainty estimation, and external validation.

Beyond comparing model architectures, this review demonstrates that high reported classification performance alone should not be interpreted as evidence of clinical readiness. Only 10 of the 81 included studies (12.3%) performed independent external validation, highlighting a substantial gap between algorithmic performance and robust clinical generalisability. For clinicians, model selection and adoption should therefore consider not only predictive performance but also external validity, calibration, robustness across patient populations and imaging environments, interpretability, computational feasibility, and compatibility with the intended clinical workflow. The proposed taxonomy and descriptive aggregation framework provide a structured basis for future comparative reviews while emphasising that successful clinical translation depends on robust validation, interpretability, reproducibility, workflow integration, and transparent reporting.

Future research should prioritise externally validated, interpretable, uncertainty-aware, and computationally efficient AI systems that are robust to heterogeneous clinical environments and supported by standardised evaluation and reporting practices. Future studies should clearly report dataset composition and class balance, data partitioning, external validation, clinically relevant performance metrics, uncertainty and calibration, computational requirements, and reproducibility information to facilitate reliable cross-study comparison and transparent assessment of clinical generalisability.

Appendix 1. Supplementary Study-Level Results

Tables 12, 13, and 14 provide the detailed characteristics and reported performance metrics extracted from the included studies. These tables support the descriptive synthesis presented in the main manuscript and are provided for completeness and transparency.

Table 12 Results presented by various studies that employed CNN for medical image classification

Category	Reference	Model	Dataset modality	Dataset size	Acc	Pre	Rec	F score	Spec
Leukemia	[59]	ALNett	Microscopic	10,661	1.0	NA	NA	NA	NA
Leukemia	[60]	DDRNet	Microscopic	13,247	1.0	1.0	1.0	1.0	NA
WBC	[60]	DDRNet	Microscopic	16,249	1.0	1.0	1.0	1.0	NA
Thyroid nodule	[61]	Fused CNN	Ultrasound	15,000	0.83	NA	0.82	NA	0.85
Brain disease	[62]	MDCNN	MRI	220	0.96	NA	NA	NA	NA
Brain disease	[63]	Parallel DCNN	MRI	3064	0.99	0.99	NA	NA	NA
	[63]		MRI	8798	0.99	0.99	NA	NA	NA
Brain disease	[64]	R-CNN	CT	10,800	1.0	NA	NA	NA	NA
Brain disease	[15]	SE-DenseNet	MRI	7023	0.96	0.97	0.97	0.97	NA
Brain disease	[65]	ProtoRadNet	MRI	7023	0.99	NA	NA	NA	NA
Alzheimer	[65]	ProtoRadNet	MRI	40,384	0.99	NA	NA	NA	NA
Chest	[65]	ProtoRadNet	CT	928	0.99	NA	NA	NA	NA
Thoracic	[65]	ProtoRadNet	X-ray	377,110	NA	NA	NA	0.5	NA
Lung disease	[66]	ECNN	CT	37,424	0.94	NA	0.90	NA	0.98
Lung disease	[67]	DM-CNN	X-ray	5854	0.99	0.99	0.99	0.99	NA
Retinal disease	[67]	DM-CNN	OCT	84,484	1.0	1.0	1.0	1.0	NA
Skin disease	[67]	DM-CNN	Dermoscopic	10,015	0.97	0.97	0.95	0.95	NA
Colorectal cancer	[67]	DM-CNN	Microscopic	100,000	1.0	1.0	1.0	1.0	NA
Lung disease	[68]	GPU-ResNet	X-ray	138	0.88	NA	NA	NA	NA
	[68]		X-ray	662	0.82	NA	NA	NA	NA
	[68]		X-ray	112,120	0.63	NA	NA	NA	NA
Lung disease	[18]	dual-input CNN	X-ray	5945	0.99	NA	NA	NA	NA
Lung disease	[69]	NIN-Based Ensemble	X-ray	5840	0.84	NA	0.88	0.90	0.97
Lung disease	[69]	GoogleNet-Ensemble	X-ray	5840	0.87	NA	0.89	0.91	0.95
Breast disease	[69]	NIN-Based Ensemble	Microscopic	272,494	0.95	NA	0.96	0.96	0.95
Breast disease	[69]	GoogLeNet Ensemble	Microscopic	272,492	0.97	NA	0.97	0.98	0.96
Lung disease	[70]	DTL	CT	852	0.93	0.95	0.91	NA	0.95
Lung disease	[71]	MGMADS-3	X-ray	17,439	0.97	NA	0.97	NA	0.98
	[71]		CT	10,839	0.98	NA	0.98	NA	0.98
Lung disease	[72]	Generalised CNN	Microscopic	1549	0.93	0.92	0.85	0.89	0.97
Lung disease	[73]	FP-CNN	Ultrasound	4305	0.97	NA	NA	NA	NA
Lung disease	[74]	Inception-ResNet	CT	2960	0.88	NA	0.83	0.96	NA
Lung disease	[75]	HESAM	CT	1145	0.99	NA	0.97	NA	0.99
Lung disease	[76]	Multitask Learning	CT	14,696	NA	NA	NA	0.86	NA
Lung disease	[76]	Compressed CNN	CT	14,696	NA	NA	NA	0.88	NA
Lung disease	[76]	Ensemble CNN	CT	14,696	NA	NA	NA	0.88	NA
Lung disease	[77]	MRFGRO	CT	746	0.99	NA	NA	NA	NA
	[77]		CT	2492	0.99	NA	NA	NA	NA
	[77]		CT	1110	0.96	NA	NA	NA	NA
Retinal disease	[78]	HahrNet	OCT	4954	0.96	NA	0.96	NA	0.99
Retinal disease	[79]	LACNN	OCT	84,484	0.94	NA	NA	NA	NA
Retinal disease	[80]	MSCNN	OCT	12,649	0.92	NA	0.92	NA	0.96
	[80]		OCT	108,312	1.00	NA	1.00	NA	1.00
Retinal disease	[36]	MedNet	OCT	109,309	0.91	NA	NA	NA	NA
Skin disease	[36]	MedNet	Dermoscopic	10,015	0.74	NA	NA	NA	NA
	[36]		Clinical Photo	16,577	0.78	NA	NA	NA	NA
WBC	[36]	MedNet	Microscopic	17,092	0.97	NA	NA	NA	NA

Table 12 continued

Category	Reference	Model	Dataset modality	Dataset size	Acc	Pre	Rec	F score	Spec
Breast disease	[81]	BCHisto-Net	Microscopic	1043	0.95	0.95	0.95	0.95	NA
	[81]		Microscopic	1429	0.89	0.90	0.90	0.90	NA
Breast disease	[82]	Improved-InceptionV3	Ultrasound	1656	0.91	0.85	1.0	0.92	NA
Breast disease	[17]	MSCNN	X-ray	1535	0.96	NA	0.97	NA	NA
	[17]		X-ray	112	0.96	NA	0.96	NA	NA
Breast disease	[83]	ConvNet	X-ray	580	0.98	0.98	0.99	0.98	NA
	[83]		X-ray	322	0.94	0.88	0.94	0.91	NA
Breast disease	[84]	TwinCNN (MLP)	Microscopic	7441	0.95	NA	NA	NA	NA
Breast disease	[84]	TwinCNN (RF)	X-ray	3104	0.80	NA	NA	NA	NA
Liver disease	[85]	Attention DCNN	Ultrasound	286	0.96	0.96	0.96	0.96	NA
Liver disease	[86]	MM-DCNN	CT	NR	0.96	NA	NA	NA	NA
Liver disease	[87]	3D-CNN	MRI	130	0.83	0.83	0.93	0.83	0.67
Skin disease	[88]	SDL	Dermoscopic	1279	0.92	NA	NA	NA	NA
	[88]		Dermoscopic	2750	0.90	NA	NA	NA	NA
Skin disease	[89]	S2-CBAM-B3	Dermoscopic	59,060	NA	NA	0.96	NA	0.36
	[89]		Dermoscopic	60,073	NA	NA	0.95	NA	0.48
Skin disease	[89]	SGE-B3	Dermoscopic	59,060	NA	NA	0.96	NA	0.31
	[89]		Dermoscopic	60,073	NA	NA	0.95	NA	0.57
Skin disease	[89]	Spatial-B3	Dermoscopic	59,060	NA	NA	0.96	NA	0.31
	[89]		Dermoscopic	60,073	NA	NA	0.95	NA	0.51
Skin disease	[89]	S2-MLP-B3	Dermoscopic	59,060	NA	NA	0.96	NA	0.27
	[89]		Dermoscopic	60,073	NA	NA	0.95	NA	0.52
Skin disease	[90]	DDCNN-F	Dermoscopic	5700	0.94	0.99	0.88	NA	0.98
Skin disease	[91]	Enhanced MobileNet	Dermoscopic	25,331	NA	0.87	0.88	0.87	NA
Alzheimer	[92]	DMA-HPCNet	MRI	4390	0.84	NA	0.86	NA	0.84
Alzheimer	[93]	Ensemble CNNs	MRI	1200	0.82	NA	NA	NA	NA
Schizoph	[94]	Inception-ResNet	MRI	NR	0.79	NA	0.80	NA	0.78
	[94]		MRI	NR	0.81	NA	0.70	NA	0.86
Polyp	[95]	MM-GLCM-CNN	CT	63	0.89	NA	NA	NA	NA
	[95]		CT	100	0.91	NA	NA	NA	NA
HLGD	[96]	Ensemble CNNs	MRI	219	0.77	0.66	0.75	0.68	NA
Parkinson	[97]	AAFP-DL	MRI	1512	0.90	NA	0.92	0.91	0.87
Bone	[98]	Dscint	Scintigraphy	1078	0.98	0.98	0.98	0.98	0.99
Ear disease	[99]	IncepResnet-v2	Otoendoscopic	10,544	0.92	NA	NA	NA	NA
Burn	[100]	BRGANeXt-50	Clinical Photo	94	0.99	1.0	0.99	0.99	NA

Table 13 Results presented by studies that leveraged ViT for medical image classification

Category	Reference	Model	Dataset modality	Dataset size	Acc	Pre	Rec	F score	Spec
Brain disease	[101]	Enhanced-ViT	CT	730	0.87	0.82	0.83	0.81	NA
Brain disease	[102]	DTViT	CT	12,651	1.0	0.99	1.0	1.0	1.0
Lung disease	[25]	POCFormer	Ultrasound	212	0.94	0.94	0.94	0.94	0.97
Lung disease	[103]	MIA-COV19D	CT	1934	NA	NA	NA	0.93	NA
Lung disease	[26]	IEViT	X-ray	6090	0.98	0.98	1.00	0.98	NA
	[26]		X-ray	4200	1.00	1.00	1.00	1.00	NA

Table 13 continued

Category	Reference	Model	Dataset modality	Dataset size	Acc	Pre	Rec	F score	Spec
	[26]		X-ray	15,162	0.99	0.99	0.97	0.98	NA
	[26]		X-ray	30,882	0.97	0.99	0.94	0.96	NA
Retinal disease	[27]	Swin-Poly Transformer	OCT	109,312	1.0	1.0	1.0	1.0	NA
	[27]		OCT	28,400	0.97	0.97	0.97	0.97	NA
Breast disease	[104]	B/32	Ultrasound	943	0.87	NA	NA	NA	NA
Skin disease	[105]	SkinTrans	Dermoscopy	10,015	0.99	0.94	NA	NA	NA
	[105]		Dermoscopy	1016	0.94	0.94	NA	0.94	NA
Skin disease	[106]	RF-C	Clinical Photo	2298	0.73	NA	NA	NA	NA
Skin disease	[106]	RF-CM	Clinical Photo	2298	0.80	NA	NA	NA	NA
Skin disease	[106]	RF-D	Dermoscopic	11,527	0.94	NA	NA	NA	NA
	[106]		Dermoscopy	33,569	0.82	NA	NA	0.82	NA
Skin disease	[106]	RF-CDM	Dermoscopy	1011	0.83	NA	NA	NA	NA
	[106]		Dermoscopy	10,351	0.93	0.85	0.82	0.81	0.99
Thyroid nodule	[107]	TC-ViT	Ultrasound	794	0.87	NA	0.87	0.92	0.86
Colorectal cancer	[108]	DIOR-ViT	Microscopic	120,028	0.88	NA	NA	0.84	NA
Prostate cancer	[108]	DIOR-ViT	Microscopic	45,912	0.78	NA	NA	0.66	NA
Gastric cancer	[108]	DIOR-ViT	Microscopic	114,806	0.85	NA	NA	0.79	NA
Femur fracture	[24]	ViT	X-ray	4207	NA	0.77	0.76	0.77	NA
Pancreatic cancer	[109]	SI-ViT	Microscopic	5088	0.94	0.92	0.91	0.93	NA
Diabetic	[110]	Swim Transformer	Microscopic	940	0.89	0.92	0.92	0.92	NA

Table 14 Results presented by studies that employed hybrid CNN-ViT for medical image classification

Category	Reference	Model	Dataset modality	Dataset size	Acc	Pre	Rec	F score	Spec
Breast disease	[111]	Eff-CTNet	Ultrasound	780	0.93	0.93	0.92	0.93	NA
Lung disease	[111]	Eff-CTNet	CT	746	0.93	0.91	0.95	0.93	NA
Lung disease	[111]	Eff-CTNet	CT	746	0.93	0.91	0.95	0.93	NA
Colorectal cancer	[111]	Eff-CTNet	Microscopic	6160	0.86	0.82	0.80	0.81	NA
Colon	[111]	Eff-CTNet	Microscopic	6160	0.86	0.82	0.80	0.81	NA
Breast disease	[112]	CViTS	Microscopic	277,524	0.95	NA	NA	NA	NA
Colorectal cancer	[112]	CViTS	Microscopic	6160	0.96	0.96	0.96	0.96	0.99
Lymphoma	[112]	CViTS	Microscopic	374	1.00	1.00	1.00	1.00	0.99
Breast disease	[113]	GasHis-Transformer	Microscopic	1390	0.88	0.84	0.83	0.83	NA
Gastric cancer	[113]	GasHis-Transformer	Microscopic	700	0.98	0.99	0.97	0.98	NA
Colorectal cancer	[113]	GasHis-Transformer	Microscopic	800	0.98	NA	NA	NA	NA
Lymphoma	[113]	GasHis-Transformer	Microscopic	374	0.84	0.82	0.83	0.83	NA
Breast disease	[114]	VGGA-ViT	Ultrasound	974	0.89	0.91	0.91	0.91	0.86
	[114]		Ultrasound	163	0.82	0.77	0.64	0.70	0.90
Breast disease	[115]	MedVIT	Ultrasound	780	0.86	NA	NA	NA	NA
Lung disease	[115]	MedVIT	X-ray	5856	0.89	NA	NA	NA	NA
Retinal disease	[115]	MedVIT	OCT	109,309	0.92	NA	NA	NA	NA
Colon	[115]	MedVIT	Microscopic	107,180	0.91	NA	NA	NA	NA
WBC	[115]	MedVIT	Microscopic	17,092	0.96	NA	NA	NA	NA
Thoracic	[115]	MedVIT	X-ray	112,120	0.76	NA	NA	NA	NA
Skin	[115]	MedVIT	Dermoscopic	10,015	0.73	NA	NA	NA	NA

Table 14 continued

Category	Reference	Model	Dataset modality	Dataset size	Acc	Pre	Rec	F score	Spec
Diabetic	[115]	MedVIT	Clinical Photo	1600	0.68	NA	NA	NA	NA
Kidney	[115]	MedVIT	Microscopic	236,386	0.67	NA	NA	NA	NA
Abdominal	[115]	MedVIT	CT	58,850	0.97	NA	NA	NA	NA
	[115]	MedVIT	CT	23,660	0.95	NA	NA	NA	NA
	[115]	MedVIT	CT	25,221	0.95	NA	NA	NA	NA
Breast disease	[116]	StoHisNet	Microscopic	4898	0.92	0.92	0.92	0.92	NA
Gastric cancer	[116]	StoHisNet	Microscopic	4290	0.95	0.95	0.95	0.95	NA
Endometrial	[116]	StoHisNet	Microscopic	3302	0.83	NA	NA	NA	NA
Lung disease	[117]	ViT-CNN	Microscopic	10,661	0.99	0.99	NA	NA	NA
Lung disease	[118]	RMT-Net	X-Ray	25,100	0.98	NA	0.98	NA	0.98
	[118]		CT	17,500	0.99	NA	0.99	NA	0.99
Lung disease	[119]	CECT	X-ray	13,808	0.98	NA	0.96	NA	0.99
	[119]		X-ray	30,386	0.97	NA	0.96	NA	0.99
Lung disease	[2]	CNN-ViT	X-ray	6432	0.99	0.98	0.98	0.98	NA
	[2]		CT	340,190	0.96	0.95	0.95	0.95	NA
Lung disease	[120]	TransUNet	CT	8474	0.85	NA	0.71	NA	0.93
Lung disease	[121]	CNN-ViT	X-ray	53,686	0.97	0.97	0.97	0.97	NA
Lung disease	[122]	MuSiC-ViT	X-ray	407,620	0.73	NA	0.64	NA	0.82
	[122]		X-ray	406,430	0.61	NA	0.30	NA	0.93
	[122]		X-ray	406,533	0.75	NA	0.59	NA	0.90
Retinal disease	[123]	HTC-Retina	OCT	48,574	0.99	0.99	0.99	0.99	1.0
	[123]		OCT	3231	1.0	1.0	1.0	1.0	1.0
	[123]		OCT	24,000	0.97	0.97	0.97	0.97	1.0
Cervical cancer	[124]	CVM-Cervix	Microscopic	8838	0.93	0.94	0.94	0.94	NA
	[124]		Microscopic	912	0.92	0.89	0.94	0.91	NA
WBC	[124]	CVM-Cervix	Microscopic	17,092	0.98	0.99	0.98	0.99	NA
Cervical cancer	[125]	CNN-ViT	Microscopic	917	0.95	0.94	0.92	0.93	NA
	[125]		Microscopic	4049	0.96	0.96	0.96	0.96	NA
	[125]		Microscopic	4789	0.85	0.86	0.85	0.85	NA
WBC	[125]	CNN-ViT	Microscopic	12,444	0.99	0.99	0.99	0.99	NA
Bone tumor	[126]	VGG16-ViT	CT	786	0.98	NA	0.93	NA	0.91
Glomerular	[127]	CVT	Clinical Photo	48,000	0.94	0.94	0.96	0.94	0.96
Tongue condition	[30]	CTransCNN	Clinical Photo	1019	NA	0.68	0.65	0.64	NA
Diabetic retinopathy	[128]	VGG16-ViT	Clinical Photo	35,126	0.82	0.79	0.82	0.78	0.63
	[128]		Clinical Photo	5590	0.84	0.84	0.84	0.84	NA
	[128]		Clinical Photo	715	0.85	0.85	0.85	0.85	NA
Liver	[129]	CMT-Net	CT	3000	0.89	0.89	0.89	0.89	NA
	[129]		CT	3004	0.89	0.89	0.89	0.89	NA
	[129]		CT	3004	0.87	0.87	0.87	0.86	NA

Appendix 2. Review-Specific Methodological Quality Assessment Framework

The criteria used for the review-specific methodological quality assessment are summarised in Table 15. These criteria were applied descriptively to identify recurring methodological strengths, limitations, and potential sources of bias across the included studies.

Table 15 Review-specific methodological quality assessment framework

Assessment domain	Criterion considered
Dataset transparency	Dataset source, size, and relevant characteristics adequately described
Preprocessing	Image preprocessing and data preparation procedures sufficiently reported
Data partitioning	Train, validation, and test separation clearly described, with attention to potential data leakage
External validation	Evaluation performed on an independent dataset, institution, or population where reported
Performance reporting	Relevant evaluation metrics and experimental results reported with sufficient clarity for interpretation
Explainability	Use and reporting of explainability or interpretability methods where applicable
Reproducibility	Availability of implementation details, datasets, code, model configuration, or other information supporting reproducibility where reported
Clinical generalisability	Evidence relevant to dataset diversity, multi-centre evaluation, domain shift, and applicability beyond the development dataset

Appendix 3. PRISMA 2020 Checklist

Table 16 presents the completed PRISMA 2020 checklist used to guide the reporting of this systematic review.

Table 16 PRISMA 2020 reporting checklist

Item	PRISMA 2020 reporting item	Location
1	Title identifies the report as a review	Title
2	Structured abstract summarising objectives, methods, results, and conclusions	"Abstract" section
3	Rationale for the review	"Introduction" section
4	Explicit review objectives and research questions	"Introduction" section
5	Eligibility criteria	"Search Strategy, Study Selection, and Quality Assessment" section
6	Information sources	"Search Strategy, Study Selection, and Quality Assessment" section; Table 2
7	Search strategy	"Search Strategy, Study Selection, and Quality Assessment" section; Table 2
8	Selection process	"Search Strategy, Study Selection, and Quality Assessment" section; Fig. 2
9	Data collection process	"Search Strategy, Study Selection, and Quality Assessment" section
10	Data items extracted	"Reporting of Results in Existing Literature" section; Tables 12–14
11	Risk of bias/quality assessment	"Search Strategy, Study Selection, and Quality Assessment" section
12	Effect measures	"Reporting of Results in Existing Literature" and "Descriptive Cross–Model Comparison" sections
13	Synthesis methods	"Search Strategy, Study Selection, and Quality Assessment" and "Descriptive Cross-Model Comparison" sections
14	Reporting bias assessment	"Limitations of the Study" section
15	Certainty assessment	Not applicable (descriptive quantitative aggregation; no formal meta-analysis)
16	Study selection	"Search Strategy, Study Selection, and Quality Assessment" section; Fig. 2
17	Characteristics of included studies	"Reporting of Results in Existing Literature" section; Tables 12–14
18	Risk of bias results	"Search Strategy, Study Selection, and Quality Assessment" section
19	Results of individual studies	Tables 12–14
20	Results of syntheses	"Reporting of Results in Existing Literature" and "Discussion and Key Insights" sections; Tables 3–11

Table 16 continued

Item	PRISMA 2020 reporting item	Location
21	Reporting biases	"Limitations of the Study" section
22	Certainty of evidence	Not applicable (descriptive quantitative aggregation)
23	Discussion	"Discussion and Key Insights" and "Future Research Directions" section
24	Limitations of the review	"Limitations of the Study" section
25	Funding	Title page
26	Data availability	Page 33
27	Registration	Not applicable

Author Contributions Hamza Hussaini: conceptualisation, methodology, literature search, data curation, formal analysis, investigation, writing—original draft preparation. Shahana Bano: methodology, writing (review and editing), supervision. Eyad Elyan: methodology, writing (review and editing), supervision. Carlos Francisco Moreno-Garcia: conceptualisation, methodology, writing (review and editing), supervision. All authors read and approved the final manuscript.

Funding The authors declare that no funds, grants, or other support were received during the preparation of this manuscript.

Data Availability The data supporting this review are derived from the published studies cited in the reference section. Study-level characteristics and extracted performance information used in the qualitative and descriptive quantitative syntheses are provided in Supplementary Tables 12–14. No new primary datasets were generated or collected as part of this review.

Declarations

Ethics Approval This article does not contain any studies with human participants or animals performed by any of the authors. Therefore, ethical approval was not required.

Consent to Participate Not applicable. This study does not involve human participants.

Consent to Publication Not applicable. This study does not contain any individual person's data in any form.

Competing Interests The authors declare no competing interests.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

References

1. Chen, X., Wang, X., Zhang, K., Fung, K.-M., Thai, T.C., Moore, K., Mannel, R.S., Liu, H., Zheng, B., Qiu, Y.: Recent advances and clinical applications of deep learning in medical image analysis. Medical image analysis. **79**, 102444 (2022)
2. Nguyen, T.-T., Nguyen, T.V., Tran, M.-T.: Collaborative consultation doctors model: Unifying cnn and vit for covid-19 diagnostic. IEEE Access. **11**, 95346–95357 (2023)
3. Yu, H., Yang, L.T., Zhang, Q., Armstrong, D., Deen, M.J.: Convolutional neural networks for medical image analysis: state-of-the-art, comparisons, improvement and perspectives. Neurocomputing. **444**, 92–110 (2021)
4. Khan, A., Rauf, Z., Sohail, A., Khan, A.R., Asif, H., Asif, A., Farooq, U.: A survey of the vision transformers and their cnn-transformer based variants. Artificial Intelligence Review. **56**(Suppl 3), 2917–2970 (2023)
5. Shamshad, F., Khan, S., Zamir, S.W., Khan, M.H., Hayat, M., Khan, F.S., Fu, H.: Transformers in medical imaging: A survey. Medical image analysis. **88**, 102802 (2023)
6. Takahashi, S., Sakaguchi, Y., Kouno, N., Takasawa, K., Ishizu, K., Akagi, Y., Aoyama, R., Teraya, N., Bolatkan, A., Shinkai, N.: Comparison of vision transformers and convolutional neural networks in medical image analysis: a systematic review. Journal of Medical Systems. **48**(1), 84 (2024)
7. Chen, C., Isa, N.A.M., Liu, X.: A review of convolutional neural network based methods for medical image classification. Computers in biology and medicine. **185**, 109507 (2025)
8. Kim, J.W., Khan, A.U., Banerjee, I.: Systematic review of hybrid vision transformer architectures for radiological image analysis. Journal of Imaging Informatics in Medicine, 1–15 (2025)
9. Page, M.J., McKenzie, J.E., Bossuyt, P.M., Boutron, I., Hoffmann, T.C., Mulrow, C.D., Shamseer, L., Tetzlaff, J.M., Akl, E.A., Brennan, S.E., et al.: The prisma 2020 statement: an updated guideline for reporting systematic reviews. bmj. **372** (2021)
10. Page, M.J., Moher, D., Bossuyt, P.M., *et al.*: Prisma 2020 explanation and elaboration: updated guidance and exemplars for reporting systematic reviews. BMJ. **372**, 160 (2021) https://doi.org/10.1136/bmj.n160
11. Whiting, P.F., Rutjes, A.W.S., Westwood, M.E., Mallett, S., Deeks, J.J., Reitsma, J.B., Leeflang, M.M.G., Sterne, J.A.C., Bossuyt, P.M.M.: Quadas-2: A revised tool for the quality assessment of diagnostic accuracy studies. Annals of Internal Medicine. **155**(8), 529–536 (2011) https://doi.org/10.7326/0003-4819-155-8-201110180-00009
12. Wolff, R.F., Moons, K.G.M., Riley, R.D., *et al.*: Probast: A tool to assess the risk of bias and applicability of prediction model studies. Annals of Internal Medicine. **170**(1), 51–58 (2019) https://doi.org/10.7326/M18-1376

13. Rahman, M.M., Munir, M., Marculescu, R.: Emcad: Efficient multi-scale convolutional attention decoding for medical image segmentation. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 11769–11779 (2024)
14. An, F., Li, X., Ma, X.: Medical image classification algorithm based on visual attention mechanism-mcnn. Oxidative Medicine and Cellular Longevity. **2021**(1), 6280690 (2021)
15. Mao, Y., Kim, J., Podina, L., Kohandel, M.: Dilated se-densenet for brain tumor mri classification. Scientific Reports. **15**(1), 3596 (2025)
16. Zhang, R., Han, X., Lei, Z., Jiang, C., Gul, I., Hu, Q., Zhai, S., Liu, H., Lian, L., Liu, Y.: Rcmnet: A deep learning model assists car-t therapy for leukemia. Computers in Biology and Medicine. **150**, 106084 (2022)
17. Niu, J., Li, H., Zhang, C., Li, D.: Multi-scale attention-based convolutional neural network for classification of breast masses in mammograms. Medical physics. **48**(7), 3878–3892 (2021)
18. Połap, D., Jaszcz, A.: Decentralized medical image classification system using dual-input cnn enhanced by spatial attention and heuristic support. Expert Systems with Applications, 124343 (2024)
19. Aggarwal, C.C.: Neural networks and deep learning. Springer. **10**(978), 3 (2018)
20. Woo, S., Park, J., Lee, J.-Y., Kweon, I.S.: Cbam: Convolutional block attention module. In: Proceedings of the European Conference on Computer Vision (ECCV), pp. 3–19 (2018)
21. Niu, Y., *et al.*: Multi-scale convolutional neural network for breast mass classification in mammograms. Computer Methods and Programs in Biomedicine. **200**, 105813 (2021)
22. Rahman, M.M., Marculescu, D.: Medical image segmentation via cascaded attention decoding. In: Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) (2023)
23. Vaswani, A.: Attention is all you need. Advances in Neural Information Processing Systems. (2017)
24. Tanzi, L., Audisio, A., Cirrincione, G., Aprato, A., Vezzetti, E.: Vision transformer for femur fracture classification. Injury. **53**(7), 2625–2634 (2022)
25. Perera, S., Adhikari, S., Yilmaz, A.: Pocformer: A lightweight transformer architecture for detection of covid-19 using point of care ultrasound. In: 2021 IEEE International Conference on Image Processing (ICIP), pp. 195–199 (2021). IEEE
26. Okolo, G.I., Katsigiannis, S., Ramzan, N.: Ievit: An enhanced vision transformer architecture for chest x-ray image classification. Computer Methods and Programs in Biomedicine. **226**, 107141 (2022)
27. He, J., Wang, J., Han, Z., Ma, J., Wang, C., Qi, M.: An interpretable transformer network for the retinal disease classification using optical coherence tomography. Scientific Reports. **13**(1), 3637 (2023)
28. Kim, J.W., Khan, A.U., Banerjee, I.: Systematic review of hybrid vision transformer architectures for radiological image analysis. Journal of Digital Imaging. **38**(5), 3248–3262 (2025) https://doi.org/10.1007/s10278-024-01322-4
29. Chen, W., Ayoub, M., Liao, M., Shi, R.: A fusion of vgg-16 and vit models for improving bone tumor classification in computed tomography. Journal of Bone Oncology. **43**, 100508 (2023) https://doi.org/10.1016/j.jbo.2023.100508
30. Wu, X., Feng, Y., Xu, H., Lin, Z., Chen, T., Li, S., Qiu, S., Liu, Q., Ma, Y., Zhang, S.: Ctranscnn: Combining transformer and cnn in multilabel medical image classification. Knowledge-Based Systems. **281**, 111030 (2023)
31. Huang, X., Ai, Z., She, C., Li, Q., Wei, Q., Xu, S., Lu, Y., Zeng, F.: A cnn-transformer fusion network for diabetic retinopathy image classification. Computerized Medical Imaging and Graphics. **126**, 102655 (2025) https://doi.org/10.1016/j.compmedimag.2025.102655
32. Dosovitskiy, A., et al.: An image is worth 16x16 words: Transformers for image recognition at scale. International Conference on Learning Representations. (2021)
33. Khan, S., et al.: Transformers in vision: A survey. ACM Computing Surveys. (2022)
34. Kelly, C.J., Karthikesalingam, A., Suleyman, M., Corrado, G., King, D.: Key challenges for delivering clinical impact with artificial intelligence. BMC Medicine. **17**(1), 195 (2019) https://doi.org/10.1186/s12916-019-1426-2
35. Lekadir, K., Feragen, A., Fofanah, A.J., *et al.*: FUTURE-AI: International consensus guideline for trustworthy and deployable artificial intelligence in healthcare. BMJ. **388**, 081554 (2025) https://doi.org/10.1136/bmj-2024-081554
36. Ferdous, M., Mahmud, S., Shimul, M.E.Z.: Mednet: a lightweight attention-augmented cnn for medical image classification. Scientific Reports. **15**(1), 41936 (2025)
37. Vasey, B., Nagendran, M., Campbell, B., *et al.*: Reporting guideline for the early-stage clinical evaluation of decision support systems driven by artificial intelligence: DECIDE-AI. Nature Medicine. **28**, 924–933 (2022) https://doi.org/10.1038/s41591-022-01772-9
38. Rivera, S.C., Liu, X., Chan, A.-W., Denniston, A.K., Calvert, M.J.: Guidelines for clinical trial protocols for interventions involving artificial intelligence: The SPIRIT-AI extension. BMJ. **370**, 3210 (2020) https://doi.org/10.1136/bmj.m3210
39. Liu, X., Rivera, S.C., Moher, D., *et al.*: Reporting guidelines for clinical trial reports for interventions involving artificial intelligence: The CONSORT-AI extension. The Lancet Digital Health. **2**(10), 537–548 (2020) https://doi.org/10.1016/S2589-7500(20)30218-1
40. U.S. Food and Drug Administration: Artificial Intelligence in Software as a Medical Device. Digital Health Center of Excellence (2025)
41. U.S. Food and Drug Administration: Premarket Notification 510(k) (2024)
42. U.S. Food and Drug Administration: De Novo Classification Request (2025)
43. U.S. Food and Drug Administration: Artificial Intelligence-Enabled Device Software Functions: Lifecycle Management and Marketing Submission Recommendations. Draft Guidance for Industry and Food and Drug Administration Staff (2025)
44. U.S. Food and Drug Administration: Good Machine Learning Practice for Medical Device Development: Guiding Principles (2025)
45. U.S. Food and Drug Administration: Marketing Submission Recommendations for a Predetermined Change Control Plan for Artificial Intelligence-Enabled Device Software Functions. Guidance for Industry and Food and Drug Administration Staff (2025)
46. U.S. Food and Drug Administration: Cybersecurity in Medical Devices: Quality Management System Considerations and Content of Premarket Submissions. Guidance for Industry and Food and Drug Administration Staff (2026)
47. World Health Organization: Ethics and Governance of Artificial Intelligence for Health: WHO Guidance. World Health Organization, Geneva (2021)
48. Sheller, M.J., Edwards, B., Reina, G.A., Martin, J., Pati, S., Kotrotsou, A., Milchenko, M., Xu, W., Marcus, D., Colen, R.R., Bakas, S.: Federated learning in medicine: Facilitating multi-institutional collaborations without sharing patient data. Scientific Reports. **10**, 12598 (2020) https://doi.org/10.1038/s41598-020-69250-1
49. Dayan, I., Roth, H.R., Zhong, A., *et al.*: Federated learning for predicting clinical outcomes in patients with COVID-19. Nature Medicine. **27**, 1735–1743 (2021) https://doi.org/10.1038/s41591-021-01506-3

50. Roberts, M., Driggs, D., Thorpe, M., Gilbey, J., Yeung, M., Ursprung, S., Aviles-Rivero, A.I., Etmann, C., McCague, C., Beer, L., *et al.*: Common pitfalls and recommendations for using machine learning to detect and prognosticate for covid-19 using chest radiographs and ct scans. Nature Machine Intelligence. **3**(3), 199–217 (2021) https://doi.org/10.1038/s42256-021-00307-0
51. Wynants, L., Van Calster, B., Collins, G.S., Riley, R.D., Heinze, G., Schuit, E., *et al.*: Prediction models for diagnosis and prognosis of covid-19 infection: Systematic review and critical appraisal. BMJ. **369**, 1328 (2020) https://doi.org/10.1136/bmj.m1328
52. Selvaraju, R.R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., Batra, D.: Grad-cam: Visual explanations from deep networks via gradient-based localization. In: Proceedings of the IEEE International Conference on Computer Vision, pp. 618–626 (2017). https://doi.org/10.1109/ICCV.2017.74
53. Ribeiro, M.T., Singh, S., Guestrin, C.: Why should i trust you?: Explaining the predictions of any classifier. In: Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pp. 1135–1144 (2016). https://doi.org/10.1145/2939672.2939778
54. Lundberg, S.M., Lee, S.-I.: A unified approach to interpreting model predictions. In: Advances in Neural Information Processing Systems, vol. 30, pp. 4765–4774 (2017)
55. Jain, S., Wallace, B.C.: Attention is not explanation. In: Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, vol. 1, pp. 3543–3556 (2019). https://doi.org/10.18653/v1/N19-1357
56. Raghu, M., Zhang, C., Kleinberg, J., Bengio, S.: Transfusion: Understanding transfer learning for medical imaging. In: Advances in Neural Information Processing Systems, vol. 32 (2019)
57. Azizi, S., Mustafa, B., Ryan, F., Beaver, Z., Freyberg, J., et al.: Big self-supervised models advance medical image classification. Proceedings of the IEEE/CVF International Conference on Computer Vision, 3478–3488 (2021)
58. Moor, M., Banerjee, O., Abad, Z.S., *et al.*: Foundation models for generalist medical artificial intelligence. Nature. **616**, 259–265 (2023) https://doi.org/10.1038/s41586-023-05881-4
59. Jawahar, M., Sharen, H., Gandomi, A.H.: Alnett: A cluster layer deep convolutional neural network for acute lymphoblastic leukemia classification. Computers in Biology and Medicine. **148**, 105894 (2022)
60. Jawahar, M., Anbarasi, L.J., Narayanan, S., Gandomi, A.H.: An attention-based deep learning for acute lymphoblastic leukemia classification. Scientific Reports. **14**(1), 17447 (2024)
61. Ma, J., Wu, F., Zhu, J., Xu, D., Kong, D.: A pre-trained convolutional neural network based method for thyroid nodule diagnosis. Ultrasonics. **73**, 221–230 (2017)
62. Hemanth, D.J., Anitha, J., Naaji, A., Geman, O., Popescu, D.E., Son, L.H.: A modified deep convolutional neural network for abnormal brain image classification. IEEE Access. **7**, 4275–4283 (2018)
63. Kazemi, A., Shiri, M.E., Sheikhahmadi, A.: Classifying tumor brain images using parallel deep learning algorithms. Computers in Biology and Medicine. **148**, 105775 (2022)
64. Uyar, K., Taşdemir, Ş., Ülker, E., Öztürk, M., Kasap, H.: Multi-class brain normality and abnormality diagnosis using modified faster r-cnn. International journal of medical informatics. **155**, 104576 (2021)
65. Sarangi, P., Agarwal, R., Basu, T.: Protoradnet: Prototypical patches of convolutional features for radiology image classification network. Artificial Intelligence in Medicine, 103324 (2025)
66. Bermejo-Peláez, D., Ash, S.Y., Washko, G.R., San José Estépar, R., Ledesma-Carbayo, M.J.: Classification of interstitial lung abnormality patterns with an ensemble of deep convolutional neural networks. Scientific reports. **10**(1), 338 (2020)
67. Han, Q., Qian, X., Xu, H., Wu, K., Meng, L., Qiu, Z., Weng, T., Zhou, B., Gao, X.: Dm-cnn: Dynamic multi-scale convolutional neural network with uncertainty quantification for medical image classification. Computers in Biology and Medicine. **168**, 107758 (2024)
68. Zhang, Q., Bai, C., Liu, Z., Yang, L.T., Yu, H., Zhao, J., Yuan, H.: A gpu-based residual network for medical image classification in smart medicine. Information Sciences. **536**, 91–100 (2020)
69. Yang, Y., Hu, Y., Zhang, X., Wang, S.: Two-stage selective ensemble of cnn via deep tree training for medical image classification. IEEE Transactions on Cybernetics. **52**(9), 9194–9207 (2021)
70. Pathak, Y., Shukla, P.K., Tiwari, A., Stalin, S., Singh, S.: Deep transfer learning based classification model for covid-19 disease. Irbm. **43**(2), 87–92 (2022)
71. Hong, G., Chen, X., Chen, J., Zhang, M., Ren, Y., Zhang, X.: A multi-scale gated multi-head attention depthwise separable cnn model for recognizing covid-19. Scientific Reports. **11**(1), 18048 (2021)
72. Momeny, M., Neshat, A.A., Gholizadeh, A., Jafarnezhad, A., Rahmanzadeh, E., Marhamati, M., Moradi, B., Ghafoorifar, A., Zhang, Y.-D.: Greedy autoaugment for classification of mycobacterium tuberculosis image via generalized deep cnn using mixed pooling based on minimum square rough entropy. Computers in Biology and Medicine. **141**, 105175 (2022)
73. Hasan, M.M., Hossain, M.M., Rahman, M.M., Azad, A., Alyami, S.A., Moni, M.A.: Fp-cnn: Fuzzy pooling-based convolutional neural network for lung ultrasound image classification with explainable ai. Computers in Biology and Medicine. **165**, 107407 (2023)
74. Chen, Y., Lin, Y., Xu, X., Ding, J., Li, C., Zeng, Y., Liu, W., Xie, W., Huang, J.: Classification of lungs infected covid-19 images based on inception-resnet. Computer methods and programs in biomedicine. **225**, 107053 (2022)
75. Lei, Y., Tian, Y., Shan, H., Zhang, J., Wang, G., Kalra, M.K.: Shape and margin-aware lung nodule classification in low-dose ct images via soft activation mapping. Medical Image Analysis. **60**, 101628 (2020)
76. Christodoulidis, S., Anthimopoulos, M., Ebner, L., Christe, A., Mougiakakou, S.: Multisource transfer learning with convolutional neural networks for lung pattern analysis. IEEE journal of biomedical and health informatics. **21**(1), 76–84 (2016)
77. Dey, A., Chattopadhyay, S., Singh, P.K., Ahmadian, A., Ferrara, M., Senu, N., Sarkar, R.: Mrfgro: a hybrid meta-heuristic feature selection method for screening covid-19 using deep features. Scientific reports. **11**(1), 24065 (2021)
78. Zhou, Q., Guo, J., Chen, Z., Chen, W., Deng, C., Yu, T., Li, F., Yan, X., Hu, T., Wang, L.: Deep learning-based classification of the anterior chamber angle in glaucoma gonioscopy. Biomedical Optics Express. **13**(9), 4668–4683 (2022)
79. Fang, L., Wang, C., Li, S., Rabbani, H., Chen, X., Liu, Z.: Attention to lesion: Lesion-aware convolutional neural network for retinal optical coherence tomography image classification. IEEE transactions on medical imaging. **38**(8), 1959–1970 (2019)
80. Sotoudeh-Paima, S., Jodeiri, A., Hajizadeh, F., Soltanian-Zadeh, H.: Multi-scale convolutional neural network for automated amd classification using retinal oct images. Computers in biology and medicine. **144**, 105368 (2022)
81. Rashmi, R., Prasad, K., Udupa, C.B.K.: Bchisto-net: Breast histopathological image classification by global and local feature aggregation. Artificial Intelligence in Medicine. **121**, 102191 (2021)
82. Sirjani, N., Oghli, M.G., Tarzamni, M.K., Gity, M., Shabanzadeh, A., Ghaderi, P., Shiri, I., Akhavan, A., Faraji, M., Taghipour, M.: A novel deep learning model for breast lesion classification using

ultrasound images: A multicenter data evaluation. Physica Medica. **107**, 102560 (2023)
83. Mahmood, T., Li, J., Pei, Y., Akhtar, F., Rehman, M.U., Wasti, S.H.: Breast lesions classifications of mammographic images using a deep convolutional neural network-based approach. Plos one. **17**(1), 0263126 (2022)
84. Oyelade, O.N., Irunokhai, E.A., Wang, H.: A twin convolutional neural network with hybrid binary optimizer for multimodal breast cancer digital image classification. Scientific Reports. **14**(1), 692 (2024)
85. Feng, X., Chen, X., Dong, C., Liu, Y., Liu, Z., Ding, R., Huang, Q.: Multi-scale information with attention integration for classification of liver fibrosis in b-mode us image. Computer Methods and Programs in Biomedicine. **215**, 106598 (2022)
86. Khan, R.A., Fu, M., Burbridge, B., Luo, Y., Wu, F.-X.: A multimodal deep neural network for multi-class liver cancer diagnosis. Neural Networks. **165**, 553–561 (2023)
87. Trivizakis, E., Manikis, G.C., Nikiforaki, K., Drevelegas, K., Constantinides, M., Drevelegas, A., Marias, K.: Extending 2-d convolutional neural networks to 3-d for advancing deep learning cancer classification with application to mri liver tumor differentiation. IEEE journal of biomedical and health informatics. **23**(3), 923–930 (2018)
88. Zhang, J., Xie, Y., Wu, Q., Xia, Y.: Medical image classification using synergic deep learning. Medical image analysis. **54**, 10–19 (2019)
89. Nazari, S., Garcia, R.: Going smaller: Attention-based models for automated melanoma diagnosis. Computers in Biology and Medicine. **185**, 109492 (2025)
90. Veeramani, N., Jayaraman, P., Krishankumar, R., Ravichandran, K.S., Gandomi, A.H.: Ddcnn-f: double decker convolutional neural network'f'feature fusion as a medical image classification framework. Scientific Reports. **14**(1), 676 (2024)
91. Cheng, H., Lian, J., Jiao, W.: Enhanced mobilenet for skin cancer image classification with fused spatial channel attention mechanism. Scientific Reports. **14**(1), 28850 (2024)
92. Mu, S., Shan, S., Li, L., Jing, S., Li, R., Zheng, C., Cui, X.: Dma-hpcnet: Dual multi-level attention hybrid pyramid convolution neural network for alzheimer's disease classification. IEEE Transactions on Neural Systems and Rehabilitation Engineering. (2024)
93. Kumar, A., Kim, J., Lyndon, D., Fulham, M., Feng, D.: An ensemble of fine-tuned convolutional neural networks for medical image classification. IEEE journal of biomedical and health informatics. **21**(1), 31–40 (2016)
94. Hu, M., Qian, X., Liu, S., Koh, A.J., Sim, K., Jiang, X., Guan, C., Zhou, J.H.: Structural and diffusion mri based schizophrenia classification using 2d pretrained and 3d naive convolutional neural networks. Schizophrenia research. **243**, 330–341 (2022)
95. Zhang, S., Wu, J., Shi, E., Yu, S., Gao, Y., Li, L.C., Kuo, L.R., Pomeroy, M.J., Liang, Z.J.: Mm-glcm-cnn: A multi-scale and multi-level based glcm-cnn for polyp classification. Computerized Medical Imaging and Graphics. **108**, 102257 (2023)
96. Mogensen, K., Guarrasi, V., Larsson, J., Hansson, W., Wåhlin, A., Koskinen, L.-O., Malm, J., Eklund, A., Soda, P., Qvarlander, S.: An optimized ensemble search approach for classification of higher-level gait disorder using brain magnetic resonance images. Computers in Biology and Medicine. **184**, 109457 (2025)
97. Cui, X., Chen, N., Zhao, C., Li, J., Zheng, X., Liu, C., Yang, J., Li, X., Yu, C., Liu, J., *et al.*: An adaptive weighted attention-enhanced deep convolutional neural network for classification of mri images of parkinson's disease. Journal of neuroscience methods. **394**, 109884 (2023)
98. Lin, Q., Cao, C., Li, T., Cao, Y., Man, Z., Wang, H.: Multiclass classification of whole-body scintigraphic images using a self-defined convolutional neural network with attention modules. Medical Physics. **48**(10), 5782–5793 (2021)
99. Cha, D., Pae, C., Seong, S.-B., Choi, J.Y., Park, H.-J.: Automated diagnosis of ear disease using ensemble deep learning with a big otoendoscopy image database. EBioMedicine. **45**, 606–614 (2019)
100. Yadav, D., Aljrees, T., Kumar, D., Kumar, A., Singh, K.U., Singh, T.: Spatial attention-based residual network for human burn identification and classification. Scientific Reports. **13**(1), 12516 (2023)
101. Ayoub, M., Liao, Z., Hussain, S., Li, L., Zhang, C.W., Wong, K.K.: End to end vision transformer architecture for brain stroke assessment based on multi-slice classification and localization using computed tomography. Computerized Medical Imaging and Graphics. **109**, 102294 (2023)
102. Fan, J., Fan, X., Song, C., Wang, X., Feng, B., Li, L., Lu, G.: Dual-task vision transformer for rapid and accurate intracerebral hemorrhage ct image classification. Scientific Reports. **14**(1), 28920 (2024)
103. Zhang, L., Wen, Y.: Mia-cov19d: a transformer-based framework for covid19 classification in chest cts. In: Proceeding of the IEEE/CVF International Conference on Computer Vision Workshops, pp. 513–8 (2021)
104. Gheflati, B., Rivaz, H.: Vision transformers for classification of breast ultrasound images. In: 2022 44th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC), pp. 480–483 (2022). IEEE
105. Xin, C., Liu, Z., Zhao, K., Miao, L., Ma, Y., Zhu, X., Zhou, Q., Wang, S., Li, L., Yang, F.: An improved transformer network for skin cancer classification. Computers in Biology and Medicine. **149**, 105939 (2022)
106. Xu, J., Huang, K., Zhong, L., Gao, Y., Sun, K., Liu, W., Zhou, Y., Guo, W., Guo, Y., Zou, Y.: Remixformer++: A multi-modal transformer model for precision skin tumor differential diagnosis with memory-efficient attention. IEEE Transactions on Medical Imaging. (2024)
107. Sun, J., Wu, B., Zhao, T., Gao, L., Xie, K., Lin, T., Sui, J., Li, X., Wu, X., Ni, X.: Classification for thyroid nodule using vit with contrastive learning in ultrasound images. Computers in biology and medicine. **152**, 106444 (2023)
108. Lee, J.C., Byeon, K., Song, B., Kim, K., Kwak, J.T.: Dior-vit: Differential ordinal learning vision transformer for cancer classification in pathology images. Medical Image Analysis. **105**, 103708 (2025)
109. Zhang, T., Feng, Y., Zhao, Y., Lei, Y., Ying, N., Song, F., He, Y., Yan, Z., Feng, Y., Yang, A.: Si-vit: Shuffle instance-based vision transformer for pancreatic cancer rose image classification. Computer Methods and Programs in Biomedicine. **244**, 107969 (2024)
110. Chen, W., Liao, D., Deng, Y., Hu, J.: Development of a transformer-based deep learning algorithm for diabetic peripheral neuropathy classification using corneal confocal microscopy images. Frontiers in Cell and Developmental Biology. **12**, 1484329 (2024)
111. Liu, S., Yue, W., Guo, Z., Wang, L.: Multi-branch cnn and grouping cascade attention for medical image classification. Scientific Reports. **14**(1), 15013 (2024)
112. Kanadath, A., Jothi, J.A.A., Urolagin, S.: Cvits-net: A cnn-vit network with skip connections for histopathology image classification. IEEE Access. (2024)
113. Chen, H., Li, C., Wang, G., Li, X., Rahaman, M.M., Sun, H., Hu, W., Li, Y., Liu, W., Sun, C.: Gashis-transformer: A multi-scale visual transformer approach for gastric histopathological image detection. Pattern Recognition. **130**, 108827 (2022)
114. Qu, X., Lu, H., Tang, W., Wang, S., Zheng, D., Hou, Y., Jiang, J.: A vgg attention vision transformer network for benign and malig-

nant classification of breast ultrasound images. Medical Physics. **49**(9), 5787–5798 (2022)
115. Manzari, O.N., Ahmadabadi, H., Kashiani, H., Shokouhi, S.B., Ayatollahi, A.: Medvit: a robust vision transformer for generalized medical image classification. Computers in Biology and Medicine. **157**, 106791 (2023)
116. Fu, B., Zhang, M., He, J., Cao, Y., Guo, Y., Wang, R.: Stohisnet: A hybrid multi-classification model with cnn and transformer for gastric pathology images. Computer Methods and Programs in Biomedicine. **221**, 106924 (2022)
117. Jiang, Z., Dong, Z., Wang, L., Jiang, W.: Method for diagnosis of acute lymphoblastic leukemia based on vit-cnn ensemble model. Computational Intelligence and Neuroscience. **2021** (2021)
118. Ren, K., Hong, G., Chen, X., Wang, Z.: A covid-19 medical image classification algorithm based on transformer. Scientific Reports. **13**(1), 5359 (2023)
119. Liu, Z., Shen, L.: Cect: Controllable ensemble cnn and transformer for covid-19 image classification. Computers in Biology and Medicine. **173**, 108388 (2024)
120. Wang, H., Zhu, H., Ding, L.: Accurate classification of lung nodules on ct images using the transunet. Frontiers in Public Health. **10**, 1060798 (2022)
121. Cao, K., Deng, T., Zhang, C., Lu, L., Li, L.: A cnn-transformer fusion network for covid-19 cxr image classification. Plos one. **17**(10), 0276758 (2022)
122. Cho, K., Kim, J., Kim, K.D., Park, S., Kim, J., Yun, J., Ahn, Y., Oh, S.Y., Lee, S.M., Seo, J.B.: Music-vit: A multi-task siamese convolutional vision transformer for differentiating change from no-change in follow-up chest radiographs. Medical Image Analysis. **89**, 102894 (2023)
123. Laouarem, A., Kara-Mohamed, C., Bourennane, E.-B., Hamdi-Cherif, A.: Htc-retina: a hybrid retinal diseases classification model using transformer-convolutional neural network from optical coherence tomography images. Computers in Biology and Medicine. **178**, 108726 (2024)
124. Liu, W., Li, C., Xu, N., Jiang, T., Rahaman, M.M., Sun, H., Wu, X., Hu, W., Chen, H., Sun, C.: Cvm-cervix: A hybrid cervical pap-smear image classification framework using cnn, visual transformer and multilayer perceptron. Pattern Recognition. **130**, 108829 (2022)
125. Fang, M., Fu, M., Liao, B., Lei, X., Wu, F.-X.: Deep integrated fusion of local and global features for cervical cell classification. Computers in Biology and Medicine. **171**, 108153 (2024)
126. Chen, W., Ayoub, M., Liao, M., Shi, R., Zhang, M., Su, F., Huang, Z., Li, Y., Wang, Y., Wong, K.K.: A fusion of vgg-16 and vit models for improving bone tumor classification in computed tomography. Journal of Bone Oncology. **43**, 100508 (2023)
127. Tian, C., Chen, Y., Liu, Y., Wang, X., Lv, Q., Li, Y., Deng, J., Liu, Y., Li, W.: Accurate classification of glomerular diseases by hyperspectral imaging and transformer. Computer Methods and Programs in Biomedicine. **254**, 108285 (2024)
128. Huang, X., Ai, Z., She, C., Li, Q., Wei, Q., Xu, S., Lu, Y., Zeng, F.: A cnn-transformer fusion network for diabetic retinopathy image classification. Computerized Medical Imaging and Graphics, 102655 (2025)
129. He, X., Wang, X., Wang, Y., Li, H., Liu, S., Wang, J., Feng, Y., Wang, Q., Chen, J.: A novel liver image classification network for accurate diagnosis of liver diseases. Scientific Reports. (2025)

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.